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INVESTIGATION OF CLINICAL MASTITIS AND CHARACTERIZATION OF ITS CAUSAL AGENTS IN DAIRY COWS IN CHATTOGRAM

Shuvo Singha

Roll No: 0118/02

Registration No.: 526

Session: 2018-2019

A thesis submitted in the complete fulfilment of the requirements for the degree of Master of Science in Epidemiology

Department of Medicine and Surgery

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Authorization

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June 2019

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This is to certify that we have examined the above Master's thesis and have found that is complete and satisfactory in all respects, and that all revisions required by the thesis examination committee have been made

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List of abbreviations

Abbreviations	Elaborations
%	*Percentages*
€	*Euro*
≥	*Greater than or equal to*
≤	*Less than or equal to*
95% CI	*95% confidence interval*
AI	*Artificial insemination*
AMR	*Antimicrobial resistance*
ANOVA	*Analysis of variance*
et al.,	*and others*
BA	*Blood agar*
BC	*Bacterial culture*
BCS	*Body condition score*
BDT	*Bangladeshi taka*
BMSCC	*Bulk milk somatic cell count*
CM	*Clinical mastitis*
CVASU	*Chattogram Veterinary and Animal Sciences University*
DCT	*Dry cow therapy*
DIM	*Days in milk*
DLS	*Department of livestock services*
IDF	*International Dairy Fedaration*
IMI	*Intramammary infection*
IRCM	*Incidence rate of clinical mastitis*
LRT	*Likelihood ratio test*
MAC	*MacConkey agar*
MALDI-TOF	*Matrix assisted laser desorption/isonization time of flight*
MIC	*Minimum inhibitory concentration*
MR test	*Methyl Red test*
MRSA	*Methicillin resistant Staphylococcus aureus*
MSA	*Mannitol salt agar*
NAS	*Non-aureus staphylococcus*
NMC	*National mastitis council*
NIR	*New infection rate*

OR	*Odd ratios*
p	*Probability*
PRTC	*Poultry Research and Training Centre*
SLU	*Swedish University of Agricultural Sciences*
SCC	*Somatic cell count*
SCM	*Sub-clinical mastitis*
SD	*Standard deviation*
SLU	*Swedish University of Agricultural Sciences*
SVA	*National Veterinary Institute, Sweden*
THI	*Temperature humidity index*
UU	*Utrecht University*
VIF	*Variance inflation factor*
VP test	*Voges-Proskauer test*
WUR	*Wageningen University and Research*

Abstract

Globally, clinical mastitis (CM) is an important production disease in the dairy industry and has a great economic impact because of reduced milk yield, milk quality deterioration, treatment costs, involuntary culling, death, increased risk of antimicrobial resistance and reduced animal welfare. A cohort study was conducted on 24 randomly selected commercial dairy farms in Chattogram for six months (from May to October 2018) to (1) estimate bulk milk somatic cell count (BMSCC) (2) estimate the incidence rate of clinical mastitis (IRCM) at different levels (Cow, quarter and farm), (3) identify risk factors for BMSCC and for CM, (4) isolate pathogens causing CM, (5) determine minimum inhibitory concentration (MIC) of selected antibiotics against *Staphylococcus aureus* and (6) describe farmers perception and behavior and treatment strategies toward CM. Clinical mastitis was defined as grade-1 if there were changes only in milk, grade-2 for changes in milk and udder and grade-3 for changes in milk and udder along with systemic changes. Farm and animal level fixed and dynamic data were recorded. Milk samples were collected from the affected animals before administering any antimicrobial therapy. The BMSCC varied from 354×10^3 to $1,157\times10^3$ cells per mL of milk during three repeated counts on the selected farms. At the start of the study, in total 1,383 lactating cows were at risk and 204 cases of CM occurred during the study period. Estimated IRCM was 43.9 (95% CI: 38.2 to 50.5%) cases per 100 cows per year where the IRCM of grade-2 (25.5 cases per 100 cows per year) was higher than the IRCM of grade-1 (10.5 cases per 100 cows per year) and grade-3 (7.8 cases per 100 cows per year). At quarter level, the IRCM was 11.6 (95% CI: 10.2-13.3%) cases per 100 quarters per year where IRCM for grade-2 was also higher (6.5 cases per 100 quarters per year) than grade-1 (2.8 cases per 100 quarters per year) and grade-3 (2.4 cases per 100 quarters per year) similarly. The IRCM varied substantially between farms, ranging from 0 cases per 100 cows per year to 103.5 cases per 100 cows per year at cow level. This also varied at quarter level between farms ranging from 0 cases per 100 quarters per year to 52.7 cases per 100 quarters per year. In generalized linear regression model, BMSCC was positively correlated with animals purchased from other than own stock and had negatively correlated with dry floor condition. Using a generalized random effects logistic regression model the following factors were determined to be

associated with higher odds of CM: high milk yield (17 liters or more per day) versus lower milk yield (OR 2.8, 95% CI 1.9-4.1); early (OR 1.6, 95% CI 1.1-6.4) and mid (OR 1.2, 95% CI 0.8-2.0) lactation compared to late lactation, and finally cows with a body condition score (BCS) ≥3.26 (OR 1.5, 95% CI 1.0-2.2) compared to a BCS ≤3. Commonly isolated pathogens from cases of CM (n=167) were non-*aureus staphylococci* (24.0%), *Streptococcus* spp. (18.6%), *Bacillus* spp. (19.8%), *Staphylococcus aureus* (3.6%), *E. coli* (6.0%) and other Gram-negative bacteria (10.2%). Of all samples, 16.3% were culture negative. One methicillin resistant *Staphylococcus aureus* was detected and found resistant against penicillin, oxacillin, cefoxitin, gentamycin and tetracycline. Altogether, the IRCM in the Chattogram region of Bangladesh is high and *Staphylococcus* spp. are the most frequently isolated pathogens. Because of the large variation in IRCM between herds and the identification of several risk factors explaining this variation, it seems feasible to improve the existing situation by better farm management practices.

Keywords*:* Bulk milk somatic cell count, incidence rate of clinical mastitis, non-*aureus staphylococcus*, matrix assisted laser desorption/isonization time of flight, minimum inhibitory concentration

Chapter-1: Introduction

In Bangladesh, livestock is one of the principal components of agro-based economy. Livestock sub-sector contributes to 20% of employment of the entire population and partial employment for another 50% of the population (Begum et al., 2011). Annual contribution of livestock is estimated as 1.5% of total Gross Domestic Product (GDP) and growth rate of GDP was 3.3% in 2016-2017 that is higher than any of the previous years starting from 2007-2008. In Bangladesh, cattle population is around 24.1 million (BER, 2017). There are some 58,590 dairy farms, each with 10 or more cows, registered with Department of Livestock Services (DLS) (The Daily Star, 2018). The annual milk production in 2017-2018 was 9,400 million litres (mL) which could ensure the milk availability of 158 mL per person per day where the target was to achieve 250 mL per day per person (DLS, 2019). Milk demand is increasing in a faster rate, but the production is insufficient. So, Bangladesh government has enforced priority in dairy development in recent years to ensure the availability of current milk deficit with good quality.

This production gap is changing in dairy farming as an emerging sector in Bangladesh. Recently, World Bank has approved USD 500 million to boost livestock and dairy production in Bangladesh (The Daily Star, 2018). But there are many obstacles in dairy sectors in this country as well. Major challenges are information gap in scientific knowledge sharing, lack of disease controlling policy, shortage of adaptable quality breeds, shortage of feed availability, insufficient training or research, lack of national milk grid or processing plants and unstable marketing (Shamsuddoha and Edwards, 2000). The livestock policy in Bangladesh is transforming local cows to cross-bred cows to increase production potential. Bangladesh is a densely populated country having shortage of grazing land for the growing dairy industry and the rearing system in Bangladesh is mostly intensive. Both increased milk production and intensive rearing system predispose dairy cows to different diseases like mastitis, reproductive problems, lameness, and different metabolic abnormalities (Britt et al., 1986; Vailes and Britt, 1990; Goldberg et al., 1992; Underwood, 1999; Phillips et al., 2013).

Mastitis is the inflammation of the mammary gland, caused mainly by infection with bacteria, and leads to changes in milk, the udder and/or the animal. The changes can be clinical or sub-clinical. Clinical mastitis is mastitis with visible symptoms, which can be mild, moderate or severe (or grade-1, 2 and 3). Mild CM (grade-1) is the physical and chemical changes in milk (milk clots, flakes, watery secretion and blood). Moderate CM (grade-2) is changes in milk together with detectable localized signs in the udder including swelling, heat, pain, or indurations of the mammary gland. And severe CM (grade-3) is defined by changes in milk and udder together with systemic signs of fever, anorexia and depression. When it cannot be determined by manual palpation or by visual examination, it is defined as sub-clinical mastitis (SCM) (Constable et al., 2017).

Bovine mastitis, whether clinical or sub-clinical, is the mostly known costly infectious disease (DeGraves and Fetrow, 1993) that results in severe economic losses from reduced milk yield, treatment costs, additional labor, milk loss during withhold treatment duration and forced culling of productive cows and drying off at an earlier lactation (Fetrow, 1980; Miller et al., 1993).

The economic loss can be minimized by early and proper diagnosis. Mastitis diagnosis may be described in two broad sections: SCM and CM detection. Identification of the burden of SCM can be determined by a direct measurement of the BMSCC with higher accuracy at herd level using somatic cell counter (Petzer et al., 2017), whereas California mastitis test (CMT) is a qualitative test, that measures the SCC indirectly and gives an estimation of SCM mainly in individual cows or quarters (Constable et al., 2017). Identification of causal agents both for CM and SCM is a gold standard for intramammary infection (IMI) detection (Constable et al., 2017) but it is not a routine practice in Bangladesh due to unavailability of technical facilities and lack of awareness of farmers and field practitioners. Identification of the causative organisms is possible through microbiological examination of collected milk sample (Pinzon-Sanchez and Ruegg, 2011; Verbeke et al., 2014; Rossi et al., 2018).

Along with diagnosis it is important to know the prevalence and associated risk factors of mastitis. Many studies have been conducted worldwide to determine the prevalence and incidence of CM, identifying various risk factors and associated

pathogen distribution. But, in Bangladesh very few studies are reported on bovine CM to measure the actual burden of clinical cases at different level (farm, animal and quarter) including risk factors association and pathogen distribution.

Clinical mastitis prevalence was reported 8.4% in Chattogram, Bangladesh (Bari et al., 2014). Data on IRCM in Bangladesh is not robust. A cohort study including 106 herds from 10 provinces in Canada reported a mean herd IRCM of 23.0 cases per 100 cow years in the selected herds ranging from 0.7 to 97.4 cases per herd (Olde Riekerink et al., 2008). Incidence rate calculated in Tanzania was 43.3 cases per 100 cows per year where quarter level IRCM was estimated 38.4 cases per 100 quarter years at risk (Kivaria et al., 2007).

Sub-clinical mastitis (15-40%) is more prevalent than CM (Sharma and Sindhu, 2007; Kumari et al., 2018), longer persistence, difficult to detect and has adverse effect on milk production. Cows with higher SCC have a positive correlation with new IMI and chronic IMI in later lactations (Lipkens et al., 2019). Confirmatory diagnosis of SCM, for example detection of BMSCC is rarely used for research purpose in Bangladesh situation. Different cut off values have been proposed for BMSCC but most studies show the acceptance of 200,000 cells per mL of milk (Barkema et al., 2013; Norman et al., 2013). High SCC and CM occurrences have been reported to be predisposed by farm level factors such as herd size, milking technique, milking machine management, poor cow hygiene and cow level factors like milk yield, lactation stage and number of parity and quarter level factors such as teat length, teat end shape, udder attachment, udder cleft, udder depth and height and width of rear udder (Deng et al., 2019; Firth et al., 2019; Miles et al., 2019; Norstebo et al., 2019). Other responsible risk factors reported for variation between the herds are breed composition, farm environment management including shed sanitation, housing hygiene and udder hygiene (washing of udder, teat disinfection and teat dipping) (Schukken et al., 1991; Elbers et al., 1998). Knowledge of risk factors is important for the implementation of the most optimal preventive measurements. Various risk factors associated with CM in Bangladesh has not been reported so far according to author's knowledge.

Mastitis is caused by a wide range of organisms. Among the organisms *Staphylococcus (S.) aureus*, coagulase negative *Staphylococci*/non-*aureus*

Staphylococci (NAS), *Streptotoccus (Str.) agalactiaae* and *Mycoplasma* are considered as contagious, whereas *Str. dysgalactiae and Str. uberis*, *Escherichia (E.) coli* and *Klebsiella* spp. are considered as major environmental organisms found in most countries. Some species of NAS are found as either be contagious or to be environmental (Watts, 1988; Bradley and Green, 1997, 2001; Bradley et al., 2007; Tiwari et al., 2013). Number of parity, type of floor materials (concrete or rubber mattress), dry cow therapy, cleaning frequency of milking machine are associated with the mastitis occurrence in presence of *Streptococcus* spp., *S.* spp. and *E. coli* (Horpiencharoen et al., 2019).

Knowing the causal agents is inevitable for proper treatment. For effective management of CM there should be precise treatment strategy. In Bangladesh there is no developed treatment strategy of CM. Antimicrobial efficacy increases success of CM treatment. But this antibiotic usage should be judicious in terms of public health safety. So, the CM is necessary to be categorized on basis of severity and involvement of pathogens to optimize the use of antibiotics. In many cases treatment is not very justifiable as milk production does not increase following treatment (McDermott et al., 1983). Other effective methods for mastitis management and control are dry cow therapy with antibiotics (Schultze, 1983), use of teat sealants in non-lactating period (Woolford et al., 1998), strip out of quarter at frequent interval, anti-inflammatory or antipyretic drugs and oxytocin for assisting milk down etc. have been described (Sato et al., 2005; Ruegg, 2017).

1.1. Objectives

So, the present study was conducted to meet following objectives:

i) To describe the BMSCC estimates over the study period

ii) To estimate IRCM at different levels of dairy cows (quarter, cow and herd) in Chattogram

iii) To know the incidence of different grading (1, 2 and 3) of CM in order to optimize therapy

iv) To identify potential risk factors in association with BMSCC and CM incidence

v) To assess the distribution of the pathogens of CM of dairy cows in Chattogram

vi) To describe farmers perception and behavior and treatment strategies toward CM

1.2. Anticipated outcome

Anticipated outcomes from the conducted research have been mentioned below

1. Estimates for BMSCC to set an standard for Bangladesh
2. Estimates of IRCM along with grades (1, 2 and 3) at individual farm, cow and quarter levels in commercial dairy farms in Chattogram
3. Determination of potential risk factors associated with BMSCC and CM in Chattogram

Chapter-2: Review of Literature

Relevant literature has been reviewed in this chapter to describe the history of dairy farming in Chattogram, the definition of CM, grading and diagnosis of CM, economic impact, CM-associated risk factors at various levels, mastitis pathogens, therapeutic management, prevention and control and antimicrobial resistance against CM pathogens. The aim of this chapter was to find out the knowledge gaps and to show the importance of conducting the present study.

2.1. Dairy farming in Bangladesh

Bangladesh is a densely populated developing country improving its economic status based on immense efforts in agricultural sub-sectors mainly through improved production in crops, livestock and fish farming (Rahman, 2017). According to Bangladesh Bureau of Statistics (BBS), 2017, the current population in Bangladesh is about 158 million which contributes 2.1% of world population (UN population division, 2019). To provide vital protein requirements for this huge population there is necessity of sustainable livestock improvement mainly in the dairy and poultry sectors. Although, from ancient time dairy farming in Bangladesh is established in a rural setting especially at the household level and with urbanization, medium and large-scale dairy farming is now becoming more common in organized form.

In total, the livestock population of Bangladesh was reported to be around 393.1 million where the cattle population was 24.1 million (DLS, 2019). The current yearly milk production is around 9,668 million litres. This was greatly expanded during the last decade as in 2007-08, the milk producing ability was around 2,729.5 million litres. Department of livestock services (DLS, 2017-2018) reported that Bangladesh has milk availability for 158 mL per person per day which is below the WHO recommended target of 250 mL per day per person. To mitigate this huge deficiency, the government is trying to give more focus on the dairy sector with the aim of increasing the number of dairy farms around the country. To boost the government's initiative the current slogan is "aamar bari aamar khamar" (my home, my farm), which was previously "one farm per house". Recently, a USD 500 million investment plan has been taken by the government to develop dairy sector along with other livestock (The Daily Star, 2018).

2.1.1. Dairy farming in Chattogram

Dairy farming in Chattogram got momentum at the beginning of 1990s' with the special initiative of the Bangladesh Government along with local elites- Yousuf Chowdhury (the then Chief Editor of The daily Newspaper "Purbakon") and Shahedul Alam Quadery, who came forward to establish their own dairy farms. Their involvement in dairying attracted many people including newly graduated persons to start dairy farming in this region. During that time, Dr. Rafiq Ahmad, a DLS officer managed to have a few breeding bulls to collect semen for developing an artificial insemination (AI) scheme to produce quality cross-bred cows for dairying in this region. This endeavor was then supported by a famous dairy practitioner Dr. Md. Farhad Hossain (currently the deputy director of DLS, Chattogram division) through offering veterinary care and needful training to dairy farmers in this area. Prof. Dr. Mohammed Shamsuddin of Bangladesh Agricultural University also contributed to develop and sustain dairy farming in Chattogram with the support of United States Department of Agriculture (USDA) funded research project. All the aforementioned activities contributed to make Chattogram become a dairy zone in Bangladesh (Personal Communication; Malik Mohammad Omar, 2019).

Nowadays, university graduates are increasingly considering dairy farming as their profession and this trend is influencing other young individuals in other parts of Bangladesh to be engaged in farming as a production opportunity of employment. In Chattogram, about 47% of the dairy farmers are doing farming as their main source of income (Khan et al., 2010). Both local indigenous and cross-bred cows are reared in different farming settings in this region. Although local breeds have adequate resistance ability against many prevalent diseases, they have a very low milk production, which makes cross-bred cows potentially more attractive (Islam, 1992). Farmers are therefore more motivated to adapt cross-bred cows instead of indigenous cows (Khan et al., 2010).

However, there are several challenges regarding the expansion of dairy farming in Bangladesh. Prevalence of infectious diseases such as mastitis, metritis, retained placenta, foot and mouth disease, hemorrhagic septicemia, anthrax, black quarter, papillomatosis etc. and non-infectious diseases such as milk fever, repeat breeding, anestrus, uterine prolapse etc. (Kabir et al., 2010; Sarker et al., 2013a), lack of disease

controlling programmes, shortage of adaptable quality cattle breeds, shortage of feed availability and high feed cost and non-organized marketing system are the major constraints in this industry (Shamsuddoha and Edwards, 2000).

2.2. Mastitis

Mastitis is a multifactorial inflammatory disease of the mammary gland that causes severe economic losses globally and it is a particularly important disease of farmers' concern for developing countries like Bangladesh.

2.3. Classification of mastitis

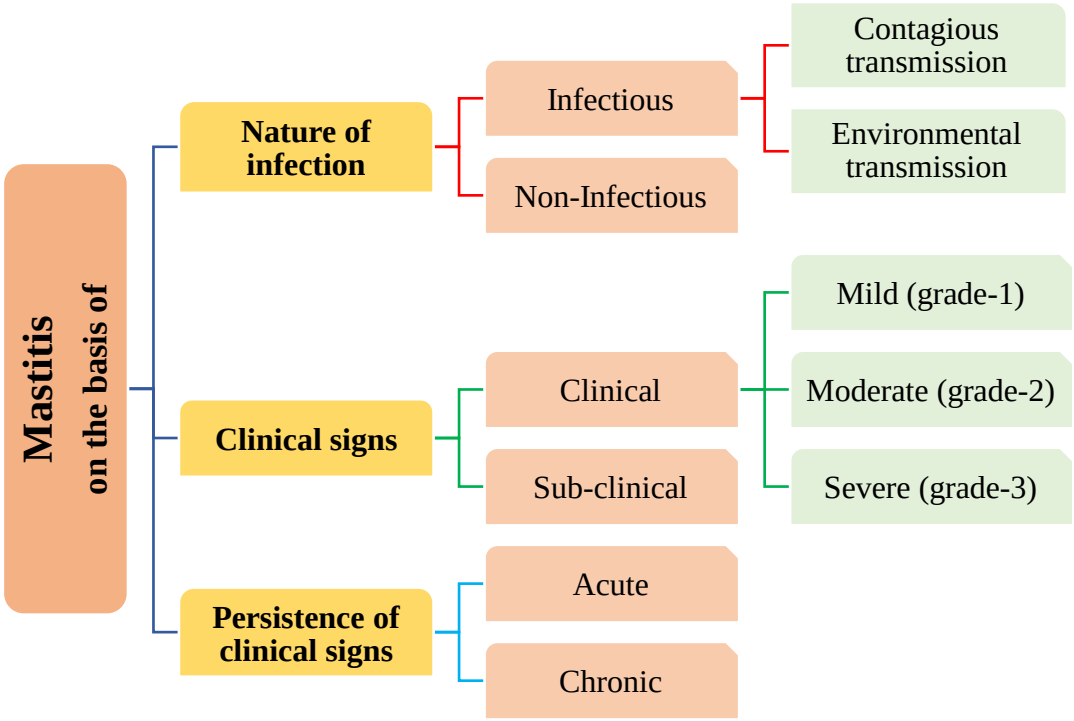

Figure 2.1. Classification of mastitis based on nature of infection, clinical signs and persistence of clinical signs

Mastitis can be classified in several ways based on nature of infection transmission, clinical signs and persistence of mastitis signs. Mastitis-causing pathogens are classified into two groups: i) contagious pathogens and ii) environmental pathogens (Bogni et al., 2011). Based on presenting clinical signs mastitis may be either clinical or sub-clinical. But SCM is difficult to detect compared to CM due to absence of any visible signs. Both contagious and environmental pathogens can adapt inside the

mammary gland and elevate somatic cell count of milk in the affected quarter which may lead to either CM or SCM.

Contagious mastitis spreads from animal to animal (generally via the milking machine or the milker), whereas environmental mastitis spreads from the environment. Moreover, in contagious mastitis the number of new cases depend on the number of infected animals in the herd, whereas in environmental mastitis, the number of new cases is independent on the number of infected animals in the herd (Nickerson, 2011; Oliver et al., 2011). Based on severity and persistence of visible signs mastitis may be either acute or chronic (Burvenich et al., 2003).

2.4. Clinical mastitis

Mastitis is recognized as clinical when pathological changes are detected either in milk (such as discoloration, presence of clots), or in the udder (such as reddening, swelling, heat and pain) or in a systemic state (such as anorexia, fever, depression and sometimes recumbency etc.) (Constable et al., 2017).

2.4.1. Grading of clinical mastitis

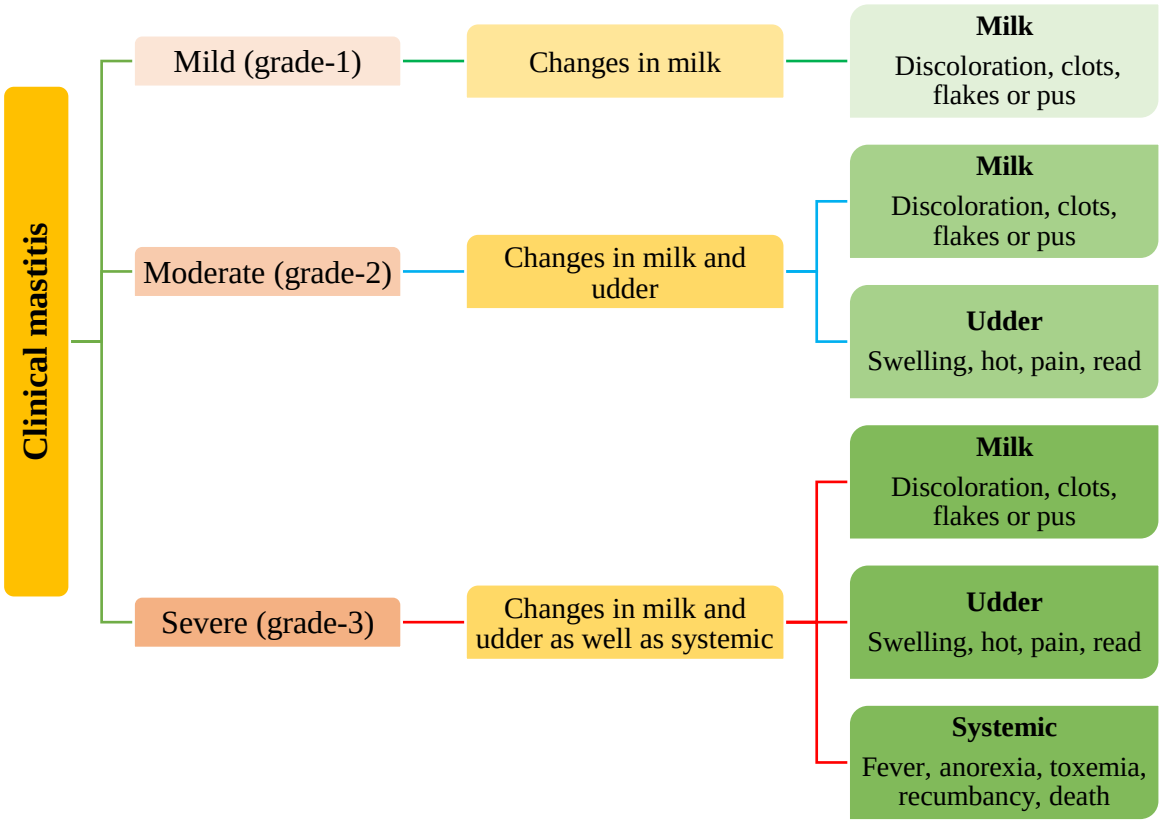

Figure 2.2. Grading of clinical mastitis

Grading based on severity of signs of CM is illustrated in Figure 2.2. Mild CM (grade -1) is denoted when limited to abnormality in milk, moderate CM (grade-2) is described as both changes in milk and udder where in the latter the changes are redness, swelling and pain, where as severe CM (grade-3) includes changes in milk and udder accompanied by systemic signs (Pinzon-Sanchez and Ruegg, 2011; Rollin et al., 2015). Milk changes are e.g. discoloration, clots, flakes or pus. Abnormalities in the udder include diffused swelling, warmth, pain or gangrene in acute cases while local fibrosis and atrophy can be found in chronic cases. Important systemic signs are degrees of anorexia, toxemia, dehydration, fever, tachycardia, ruminal stasis followed by recumbency and death (Pinzon-Sanchez and Ruegg, 2011; Constable et al., 2017).

2.5. Diagnosis of clinical mastitis

Clinical mastitis is diagnosed by looking at visible changes in milk, udder and/or systematic abnormalities of the affected animals and by conducting bacteriological culture (BC) for further confirmation. Clinical mastitis can also be detected in various approaches at the herd level or individual animal (Petzer et al., 2017). At individual cow and quarter level this may be done using e.g. California Mastitis test (CMT) and SCC of an individual cow or bulk milk for detection of CM in herd (Green et al., 2004; de Haas et al., 2005). Very high level of CMT score or SCC of individual quarter indicates mastitis. However very high BMSCC may not ensure detection of CM affected quarter but may have assumption that some cows may be affected by CM. Although a single case in herd might not have very prominent change in BMSCC. Clinical mastitis cases should be confirmed based on BC results. Bacterial culture will reveal presence of specific pathogen responsible for particular CM case (Adkins et al., 2017).

Clinical mastitis diagnosis needs to be confirmed by differentiating it from other udder abnormalities such as udder edema, rupture of suspensory ligament, hematoma and bloody milk from freshly parturiated cows (Constable et al., 2017). In udder edema anterior ventral abdomen is excessively swollen revealing pits on pressure while in rupture of ligaments are differentiated by right, left or ventral displacement of udder. Presence of blood through milk indicates the rupture of blood vessels of mammary gland through trauma. These diseases are indirectly associated with CM

cases (Tucker et al., 1992; Jackson and Cockcroft, 2002; Constable et al., 2017; Singha et al., 2019).

2.6. Milk somatic cell count

Somatic cell count of milk from individual quarter or cows represents the inflammatory status of the mammary gland (Schukken et al., 2003). Bulk milk SCC is a worldwide acceptable sensitive (88%) and specific (80%) indicator for the status of herd udder health as well as milk quality (Zecconi et al., 2019). This serves as an indirect measure for the mastitis burden at herd level. Exceeding permissible limits of BMSCC indicates further required investigation of the herd (Constable et al., 2017). Somatic cells are mostly cells of immune system and part of natural defense mechanism. The cells include lymphocyte, macrophages, polymorphonuclear cells and some epithelial cells but polymorphonuclear cells increase up to 90% with the increase SCC (Zecconi et al., 1994). Number of SCC increases when there is bacterial infection and tissue damage (Ryšánek et al., 2001). So, SCC can be used to distinguish between infected and uninfected quarters (Pillai et al., 2001). Report on BMSCC in Bangladesh is scarce due to unawareness of its use and unavailability of somatic cell counters. In international standard, the range of BMSCC below ***200,000*** cells per mL of milk is indicated as a safety margin (Barkema et al., 2013; Norman et al., 2013) where many countries of Europe consider the acceptable value of ***400,000*** cells per mL and in Canada the exceeding limit was reported to be ***500,000*** cells per mL (Barkema et al., 1998; Sargeant et al., 1998) in drinkable milk. In other countries like USA, standard legal limit for SCC in milk is ***750,000*** cells per mL (Norman et al., 2019) which varies between different states. Naturally there is a slight increase level of BMSCC at early and then again at late lactation so average of BMSCC at three lactation stages gives a reasonable value for BMSCC. Although it is not possible to use BMSCC estimates for calculating exact number of affected cows in a herd but this gives a good estimate for number of affected quarters at herd level (Constable et al., 2017).

There are no such standard for developing countries. In a recent study, BMSCC was more than ***400,000*** per mL of milk in 50% of the studied farms which we are considering as standard in Bangladesh situation (Singha et al., 2019).

2.7. Economic aspects of mastitis

Mastitis is described as the costliest disease on dairy farms around the world. This is due to a decline in milk production, discarding of milk, deterioration of milk quality, cost of veterinary services/diagnostics/drugs, additional management, intensified risk of recurrence, increased risk for culling, and additional materials for prevention of mastitis (Hogeveen et al., 2011). Worldwide economic losses due to CM is huge and may vary between farms and this is the most frequent cause of antibiotic use in dairy farms (Hoblet et al., 1991). Reduced milk production also hampers the calf weaning weights (Waller et al., 2014). The cost of CM and prevention of mastitis in dairy herds have been reported considering 9 dairy herds in Columbia over one year that concluded each CM case costs €33.5 per cow-year, with a mean cost per clinical episode was €94.6. Loss of milk production estimated €13.1 per cow per year which excludes associated losses of SCM (Hoblet et al., 1991). In Sweden, the maximum avoidable losses for CM were estimated €97 per cow-year (Hagnestam-Nielsen and Ostergaard, 2009). In the Netherlands economic losses of CM estimates ranged from €61 to €97 per cow with large differences between farms. Losses from CM case estimates to be €210 and may vary €235 at post-partum, whereas €164 at the lactation end (Hogeveen et al., 2011). Clinical mastitis is also responsible for huge loss to the dairy industry in Bangladesh and the annual economic losses due to mastitis have estimated BDT. 122.6 (€ 1.3) million (Kader et al., 2003). In the current situation the amount should be quite higher.

2.8. Incidence rate of clinical mastitis

Several studies have been undertaken to determine the IRCM in different countries except Bangladesh (Table 2.1).

Table 2.1. Reports on incidence rate of clinical mastitis from different countries

Incidence rate (100 cows per year)	Quarter level (100 quarters per year)	Country	Reference
47.0	-	UK	Bradley et al. (2007)
43.3	38.4	Tanzania	Kivaria et al. (2007)
41.6	-	UK	Bradley and Green., (2001b)
32.5	-	Netherlands	Santman-Berends et al., (2015)
31.7	-	Tanzania	Karimuribo et al.

			(2006)
26.3	-	Canada	Thompson-Crispi et al. (2013)
23.3	-	Canada	Olde Riekerink et al. (2008)
20.9	-	Canada	Levison et al. (2016)
12.7	-	Netherlands	Miltenburg et al. (1996)
-	3.7	Brazil	Tomazi et al. (2018)

The IRCM is not frequent and systematic in Bangladesh according to the author's knowledge. But there were some sporadic reports on prevalence of CM. Prevalence of CM was reported 55% in Chattogram (Jha et al., 2010). So, the core objective of our study was to illustrate the grading of CM and calculation of IRCM in Bangladesh at various levels of farm, animal and quarter.

2.9. Risk factors associated with clinical mastitis

A number of studies have been accomplished to define and estimate the effects of different factors responsible for the occurrence of CM at different levels.

2.9.1. Farm level risk factors

Farm level risk factors associated with CM are various ecological factors, seasonal factors and managemental factors which are discussed below under each sub-heading.

2.9.1.1. Ecological factors

Temperature and humidity can play an indirect role in association with production parameters and quality of milk. High temperature-humidity index (THI) is negatively correlated (r=-0.16) with milk production and positively correlated (r=0.26) with the SCC subsequently (Nasr and El-Tarabany, 2017). This may be due to extreme THI, the animals face degrees of stress which in turn have negative influence on other factors and might be associated with decreased production parameters and/or high SCC.

2.9.1.2. Seasonal factors

The odds of CM were 1.2 times higher in rainy season compared to dry season (Oliveira et al., 2015). A study in Iran reported that occurrence of CM was higher in summer which was correlated with late lactation. But, in winter, CM cases were correlated with early lactation (Moosavi et al., 2014). Another study had reported the

increased IRCM as 11.6 cases per 10,000 quarter days at risk in rainy season while 9.6 cases per 10,000 quarter days at risk in dry season (Tomazi et al., 2018). In summer there is a presence of high temperature and humidity which may put the animals into stress condition. In other direction, in rainy season there is more humidity and the wet condition may also support organisms for rapid multiplication.

2.9.1.3. Managemental factors

Cows reared in tie stalls have higher incidence rate of CM compared to cows reared in free stalls (Olde Riekerink et al., 2008). Herds having 31 or more cows was reported associated with high IRCM which was 62 cases per 100 cow per year (Kivaria et al., 2007). Apparently visible association of herd size was probably due to spatial differences between study sites. This might be dependent on level of managemental standard which vary from region to region or between farms. Generally, good udder preparation e.g., washing of teat and udders, teat sanitation using pre and post-dipping, drying using a single towel for each cow has a significant negative relationship with the mastitis occurrence (Pankey et al., 1987; Pankey, 1989; Tolosa et al., 2015). Udder of cows wiped with individual towel and feeding the cows immediately after milking was associated with a lower incidence of CM (Kivaria et al., 2007). Studies on relationship between educational qualifications and infected herds showed that 61.1% cases were from the farms owned by illiterate owners in comparison to owners just finished primary education (34.8%), owners finished secondary education (25.9%) and owners with education of collegiate standard or more (12.5%) (Suresh et al., 2017). With the upgraded level of education, farmers are able to take better managemental decisions (Pushpa, 2006).

2.9.2. Animal level risk factors

There are many animal level factors that may be indirectly associated with CM such as BCS, parity, lactation stage, milk yield and use of dry cow therapy (DCT) (Royster and Wagner, 2015; Beyene and Tolosa, 2017). Separately each of the factors has been discussed in beneath.

2.9.2.2. Breed

Bos taurus taurus (exotic breeds in Indian sub-continent and Africa are six times more associated with Non-*aureus staphylococcus* (NAS) causing CM than *Bos taurus*

indicus (native breeds in Indian sub-continent and Africa) (Oliveira et al., 2015). Another study by Pinzon-Sanchez and Ruegg, (2011) also had reported the evidence that *Bos taurus taurus* was 5 times more commonly affected with CM compared to *Bos taurus indicus.* Zebu cattle was frequently reported to show less susceptibility to CM probably due to low genetic potential for milk production (Wilson et al., 1997; Shem et al., 2002; Dego and Tareke, 2003). Holstein Friesian is an exotic cattle breed and this breed has high genetic potentiality of dairy characters which has been proved associated with susceptibility to CM (Hansen et al., 2002). This reason was not clearly reported in other studies but this could be explained as improvement in the milk yield trait is negatively related with genetic resistance trait against mastitis

2.9.2.3. Body condition score

Poor BCS was associated with a higher IRCM of 43.1 cases per 100 cow per year at risk compared to the group of fair (37.2 cases per 100 cows per year) and good BCS (33.5 cases per 100 cows per year) of lactating cows (Kivaria et al., 2007). But it is difficult to demonstrate direct relationship of BCS and CM as BCS may be associated with multiple factors of various metabolic states which may combinedly lead to CM. Further research is needed to precisely estimate the association of BCS with causation of CM.

2.9.2.4. Parity

Parity-wise IRCM shows 35.8, 56.7 and 76.7 cases per 100 cows per year at risk correspondingly for parity of 1-4, 5-8 and ≥9 (Kivaria et al., 2007). In another study, animals of 3rd parity or more (OR: 2.4 times) and, 2nd parity (OR: 1.7 times) showed a higher tendency for having CM compared to newly calved heifers (Breen et al., 2009). This might be due to the standard care are provided to the young animals and generally they are separately reared from the main herd. Moreover, older animals usually have subsequent other conditions like lameness. In older cows, there might be increased persistence of pathogens inside the udder (Breen et al., 2009). Cows having higher parity usually have loosened sphincter and patency in teat canal. Median ligaments also get remarkably relaxed which lead to hanging of udder much below than hock. So, teat canal can be injured and may predispose the entry of mastitis organisms much easily through the teat canal (Bhat et al., 2017).

2.9.2.5. Lactation stage

Multiple studies reported the occurrence of CM at various lactation stages. At first month of lactation the cows were having CM 8.1 times more compared to 2nd month or above (Oliveira et al., 2015). Days in milk in a lactation stage and average milk yield per day also play a role leading to mastitis. Cows 120 or less days in milk results in higher IRCM of 84.9 cases per 100 cows per year at risk than onwards days in milk in a lactation (Kivaria et al., 2007). Reasons behind this association may be due to occurrences of most new infections are associated with early part of the dry period and at calving. So, early lactation stage will be the most vulnerable period for the CM occurrences.

2.9.2.6. Milk yield

High milk producing trait has been documented for association with occurrences of CM. In previous studies milk yield of 20 litres were reported correlated with incidence for CM (Bhat et al., 2017; Tomazi et al., 2018). Higher milk yield may cause loss of micronutrients which in turn may play role hindering body immunity. So, the animals may easily get invaded by micro-organisms (Bhat et al., 2017). There is also some genetic correlation of milk yielding trait of cows with occurrences of CM, probably due to production stress in high milk producers.

2.9.2.7. Dry cow management

Dry cow therapy denotes infusion of antibiotics into the udder through the teat canal at the end of the lactation period. Bhutto et al., (2011) reported that there were less cases of CM in cows infused with antibiotics (Cloxacillin or Framycetin) compared to cows that were not treated in dry period (Bhutto et al., 2011). Another study also concluded use of DCT had significant negative relationship with occurrence of CM (Kivaria et al., 2007). Dry cow therapy has been proved effective against occurrence of CM in early lactation through prevention of the new intramammary infection during the dry period (Bhutto et al., 2011). Selective DCT was suggested for having an increased success rate of eliminating major mastitis causing pathogens (Østerås et al., 1999). Moreover, there is no difference in therapeutically success rates between blanket DCT and selective DCT and will optimize the costs (Rindsig et al., 1978; Cameron et al., 2015; Vanhoudt et al., 2018). As a control measure for CM, DCT may

shorten the duration of existing infection in dry period. So, DCT reduces the exposure and way of transmission and finally decrease the likelihood of CM in next lactation.

2.9.2.8. Calf suckling

Calf suckling may contribute to the invasion of mastitis causing pathogens. So, separating the cows from neonatal may minimize the risk of infection but some health benefits have previously been studied through cow-calf contact (Beaver et al., 2019). Calf sucking may help remove residual milk from udder which indirectly might help reduce risk of getting mastitis. Moreover, calf sucking may also help in bacterial inhibition through the active lysozymes from calf saliva (Mdegela et al., 2004). During suckling cows may get injury to teat through rough bites by the calf. This condition is called teat chap and this may lead to mastitis (Thomas et al., 1981; Jackson and Cockcroft, 2002). But this can be well maintained by nursing for short duration (Walsh, 1974). Calf suckling keeps the teat canal open for long time which may increase the possibility of udder infection (Personal communication: Md. Nurul Amin, 2019).

2.9.3. Cow and quarter level risk factors

Cow and quarter-level risk factors have been reported to play a major role in association with causation of mastitis in many studies. Flat teat shape were 1.5 time more contributing in developing CM compared to teats with round ends (Miles et al., 2019). Previous teat lesions have a high correlation with incidence of CM (Kivaria et al., 2007). Left fore and right hind quarter were affected more in studies in Jammu region, India (Bhat et al., 2017).

Literature was discussed above for properly describing the effects of responsible risk factors behind mastitis at farm, individual cow and quarter levels. But there were lack of information to identify the actual contributing risk factors causing CM in Bangladesh. The present study was designed to calculate the contribution of each factor for IRCM and also to measure the combined effect of certain factors to cause CM in respect of farming condition in Bangladesh.

2.10. Mastitis pathogens

Mastitis is caused by a wide variety of microorganisms. Various species of bacteria and several species of fungi are responsible for causing mastitis (Constable et al.,

2017). On epidemiological basis they can be classified into contagious and environmental pathogens (Cervinkova et al., 2013; Tiwari et al., 2013).

2.10.1. Contagious pathogens

Contagious pathogens are those pathogens that live on the udder or the cow's skin and mucosal sites or as an intramammary infection to serve as reservoir for carrying the infections to other cows in the herd (Nickerson, 2011). They may spread towards one cow to another mainly during milking. The common contagious mastitis pathogens are *Str agalactiae, S. aureus*, several non-*aureus staphylococci* (NAS), *Mycoplasma* spp., and *Corynebacterium bovis* (Verbeke et al., 2014). Multiple contagious microorganisms have been isolated from different mastitis cases worldwide. In many studies it is shown that *Staphylococcus* spp. : 11.8% to 38.3% (Cheng et al., 2019; Horpiencharoen et al., 2019) and *Str.* spp. : 18.7% (Cheng et al., 2019) are the chief contagious pathogens. The distribution of major contagious pathogens ranged as *S. aureus* : 10.2% to 44.3% (Waage et al., 1999; Gao et al., 2017), NAS : 4.5% to 19.8% (Breen et al., 2009; Oliveira et al., 2015; Cheng et al., 2019), *Str. Agalactiae* : 5.9% to 7% (Oliveira et al., 2015; Tomazi et al., 2018), and *Corynebacterium* spp. : 2.3% (Breen et al., 2009). *Str.* spp. and *Staphylococcus* spp., especially NAS, are commonly found in the teat skin or in the streak canal. So, their presence indicates presence of poor teat hygiene (Oliveira et al., 2015).

2.10.2. Environmental pathogens

Environmental pathogens are those that infect the udder from the environment. The known common pathogens are *Escherichia* (*E.) coli, Klebsiella* spp., *Str. dysgalactiae* and *Str. uberis,* and some NAS (Smith et al., 1985; Harmon, 1994). Major isolated environmental pathogens distribution for CM occurrences were reported as *E. coli* : 6.6% to 20% (Verbeke et al., 2014; Tomazi et al., 2018), *Klebsiella* spp. : 10% to 13% (Sumathi et al., 2008; Gao et al., 2017), *Pseudomonas* spp.: 0.3% (Breen et al., 2009), *Proteus* spp. : 0.3% (Breen et al., 2009), *Enterobacter aerogenes* : 0.2% (Breen et al., 2009), *Str. dysgalactiae* : 10.5% to 18.2% (Waage et al., 1999; Gao et al., 2017) and *Bacillus* spp. : 1.3% (Breen et al., 2009). The reasons of environmental pathogen distribution might be due to sudden changes of housing system (tie stall to free stall) or reduced load of *Str, Staphylococcus* or other minor pathogens (Oliveira et al., 2015). The other conditions might be wet bedding, dirty lots, milking wet

udders, inadequate preparation of teat or udders which predispose infection with environmental pathogens (Constable et al., 2017).

Very few studies have previously been reported in Bangladesh to identify the involved pathogens in CM cases. So, it was important to identify the types of pathogens associated at each affected quarter. In the present study, associated pathogens were isolated from CM cases and distribution was presented at farm, animal and quarter levels.

2.11. Mastitis management

Mastitis treatment and control guidelines have been described by National Mastitis Council (NMC) 2017 and Nordic guidelines for mastitis therapy, 2009. This follows the standard protocol for treatment and controlling strategies against mastitis.

2.11.1. Treatment strategy

The standard protocol by NMC (2017) suggests that cows with CM should not be treated with an antibiotic prior to the results from bacteriological culture results to determine the type of organisms. The samples will be tested on farm set up of laboratory or will be transported to a lab and within 24 hours the culture results will be forwarded and the choice of antibiotics will be performed accordingly. Cows infected with Gram negative bacteria (*E. coli* or *Klebsiella* spp.) will not be treated with any antibiotics, but supporting drugs will be provided as oxytocin to enhance the milk let-down, intravenous or oral fluid and non-steroid anti-inflammatory drugs (Landin et al., 2011). Confirmed Gram positive bacterial infection will be treated with an antibiotic according to results of antibiotic susceptibility testing. By this period of culturing procedure (except severe CM) for 24 hours, cows will be only treated with supportive therapy like oral or systemic fluid and anti-inflammatory drugs if animal has dehydration or fever depending on severity of clinical symptoms (Adkins et al., 2017). At present situation there is no established guideline for treating CM cases in Bangladesh. This is a major focus of current study that through knowledge of correct grading, an ideal treatment strategy will be implemented against CM cases. NMC guidelines for treatment have been attached as a figure in the **Appendix-V**.

2.11.2. Antimicrobial resistance and mastitis

Mastitis is one of the infectious diseases in dairy cows mainly treated with antibiotics worldwide. Many resistant mastitis causing bacteria has been isolated from clinical cases of bovine mastitis. Several studies have been reviewed from Bangladesh and surrounding countries on antimicrobial sensitivity status against mastitis causing pathogens. In vitro antimicrobial sensitivity test against *S. aureus* reported that sensitivity status varied among various countries such as for gentamycin (58.6% to 91.1%) (Yang et al., 2016; Hoque et al., 2018), ciprofloxacin (50.5% to 100%) (Hoque et al., 2018), enrofloxacin (63.8% to 88%), erythromycin (24% to 95%) (Saini et al., 2012; Yang et al., 2016), tetracycline (25.5% to 84%) (Saini et al., 2012; Hoque et al., 2018), neomycin (50%), cephalexin (47%), penicillin (38% to 86%) (Saini et al., 2012; Yang et al., 2016; Käppeli et al., 2019), oxacillin (2% to 100%) (Saini et al., 2012; Hoque et al., 2018; Käppeli et al., 2019), methicillin (2.3%) (Yang et al., 2016), streptomycin (25.5% to 35%), sulfadiazine (30% to 52%) (Saini et al., 2012), ampicillin (0% to 82%) (Saini et al., 2012) and amoxycillin (0 to 58.0%) (Hoque et al., 2018). This available data on antimicrobial susceptibility testing using disk diffusion method is not able to reflect the actual antimicrobial response against certain pathogens because minimum inhibitory concentration (MIC) can define actual dose response of certain pathogen. So, in this current study we wanted to see the MIC of commonly used antimicrobials against an important contagious mastitis pathogen; *S. aureus*.

2.11.3. Prevention and control

For controlling occurrence of CM there are predefined 10-point control strategies by NMC, (2017): i) Udder health goal should be setup by targeting periodic SCC and reviewing the improvement over time, ii) Stall size should be adequate and clean and comfortable environment should be maintained, iii) Standard milking procedures must be applied ensuring the pre and post-dipping of teats and disinfection of hands before milking, iv) Installation, servicing and replacement of improper milking equipment should be done based on examination, v) Good record keeping should be ensured, vi) Proper CM management will be considered during lactation, vii) Effective DCT through proper feeding plan, drying the udder, sealing of the teats with long acting antibiotics, viii) Strict maintenance of biosecurity using the BMSCC data for

suspicion and diagnostics, aseptic milk sampling, segregation or culling of infected animals based on bacteriological confirmation, ix) Regular monitoring of udder health status through enrollment in regular SCC monitoring programme, monitoring the variation in distribution of SCC values, calculating the high SCC and CM incidence rates for making treatment and marketing decisions, and x) Finally, periodic review of complete mastitis control programme will be maintained by an advisory team including veterinarian, producer, herd manager and milking personnel (NMC, 2017).

2.11.3.1. Vaccination

There are vaccines available in different countries against mastitis. Reported available vaccines were whole organism vaccines, DNA vaccines encoded by clumping factor-A, live attenuated *S. aureus*, capsular polysaccharide protein conjugated vaccines and type-C enterotoxin mutated recombinant vaccine of *S. aureus* (Poutrel et al., 1988; Leitner et al., 2003; Buzzola et al., 2006; Nour El-Din et al., 2006; Chang et al., 2008). In an experiment, *S. aureus* oil adjuvant carried inactivated vaccine was used for field trial and showed potential protective action against in whole lactation (Nordhaug et al., 1994). Another prospective cohort study was undertaken to show the efficacy of *E. coli* mutant vaccine against coliform CM. The IRCM in control group was 12.8 cases per 100 cases where in vaccinated group the rate was 2.6 cases per 100 cases (Gonzalez et al., 1989). Mastitis vaccination concept is not very well established in Bangladesh and very few of the vaccines were used commercially. The efficacy of various vaccines is questionable regarding the availability of suitable vaccine strains, in-vivo success rates for preventing CM cases and economic efficiency. The efficacy of any vaccine available in Bangladesh has not been systematically proved yet so far according to author's knowledge. There were very few reports of vaccination efficacy specifically on CM. To expand the actual reasons for efficacy of vaccination against CM there should be further research.

2.12. Conclusion

In conclusion, prevalence and incidence of CM, BMSCC distribution across herds, economic impacts of mastitis, various risk factors at farm, animal and quarter level, diagnostic approach along with prevention and control strategies and pathogens' sensitivity against antimicrobials were discussed in coherence. Knowledge gaps are the IRCM and the distribution of BMSCC, association of BMSCC with farm level

factors, isolation of associated pathogens with CM and antimicrobial resistance against isolated *S. aureus* in Bangladesh. This should be done through an appropriate prospective study design. Therefore, conducting a cohort study on bovine CM is justified to fill the aforementioned gaps.

Chapter-3: Materials and Methods

3.1. Description of the study site

Chattogram district is located at south eastern part of Bangladesh at 22º0'45"N and 92º6'1"E. This district is subdivided into 18 administrative areas called “upazillas or thanas”. The average temperature of this district ranges from 13º to 32ºC, humidity 70 to 85%. The average rainfall varies from 5.6 mm to 727.0 mm (Anon, 2016). Involvement of several elite persons (such as newspaper editors, politicians etc.) at the beginning of the 90s, government support, veterinary care and breeding support from several dedicated veterinarians in Chattogram region helped in dairy expansion in this area of Bangladesh (Personal communication: Malik Mohammad Omar, 2019). Around 0.3 million cattle heads that contribute to 1.2% of total cattle population in Bangladesh (BBS, 2011) made this district suitable to conduct the present investigation on bovine CM in selected commercial dairy farms.

3.2. Dairy farming of Chattogram

Dairy farming is a profitable business in Bangladesh perspective (Shamsuddin et al., 2007). Around 1,551 commercial dairy farms in different sizes are available in Chattogram. The animals in these farms are mainly cross-bred (between Local and Holstein Friesian; Holstein Friesian and Jersey; Holstein Friesian and Shahiwal; Holstein Friesian and Gyr). The department of livestock services, Bangladesh has been implementing a cross-breeding scheme through an AI programme dominated by Holstein Friesian, and 50-62.5% blood is considered as suitable for Bangladeshi climate (Singha et al., 2019). Average percentage of cows in lactation per farm in Chattogram has been reported to be 50 to 70% (BBS, 2011). Farm hygiene and bio-security standard of farms along with record keeping system in Chattogram district is generally poor (Chowdhury et al., 2017) and therefore dairy farming in this district has many challenges like production diseases (mastitis).

3.3. Study design, sample size and sampling

All commercial dairy farms under Chattogram district were considered as reference population for the study. A comprehensive list of 102 dairy farms, each consisting of at least 3 lactating cows, under Chattogram district was previously developed during the Swedish funded project: “Development of udder health control programme in

dairy cows in Bangladesh". Of 102 farms, 24 (From 7 upazillas and 7 thanas) were selected (Figure: 3.1) based on farmer's willingness, capacity to keep records accurately, easy access and manageable numbers for a 6-month long cohort study (May to October 2018) to investigate epidemiology of CM.

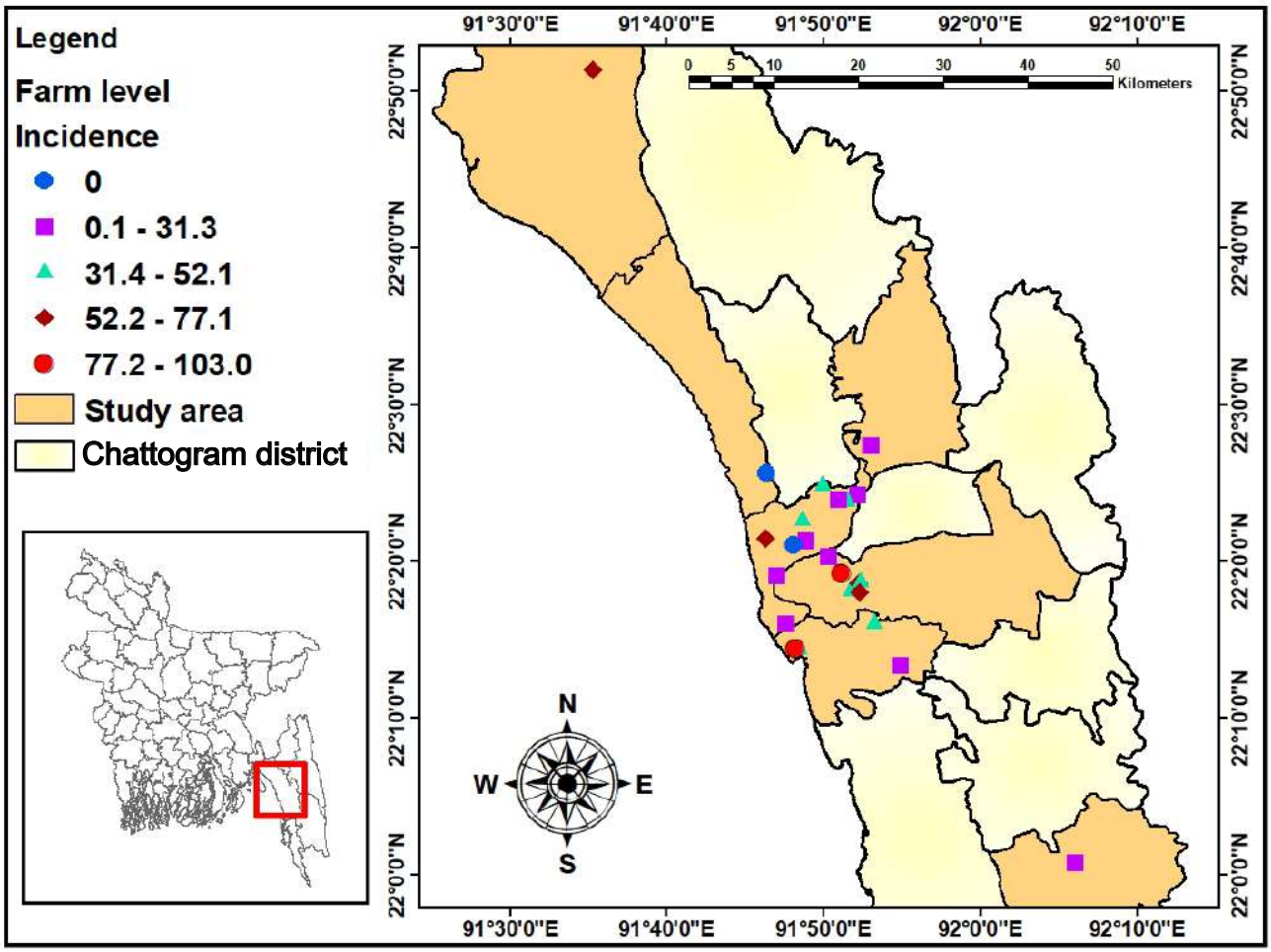

Figure 3.1. Location of selected dairy farms in Chattogram District

3.4. Epidemiological data collection

A questionnaire was developed and then pretested prior to the start of the main study. The questionnaire was designed to record 2 types of information at different levels: i) baseline data and ii) dynamic data. The questionnaires were administered either to farm owner, manager or responsible farm employee and recorded through face to face interviews and also by physical observation.

3.4.1. Baseline data collection

Baseline data (both on farm and animal level) were collected from each selected farm. Farm level data included address and location of the farm, farmer's demography, farm composition etc. Animal level data consisted of animal identity number, BCS, lactation stage, milk yield, days after parturition etc. Data related to farmers' perception on mastitis and treatment strategies were also recorded. Detailed questionnaire has been attached in **Appendix-I**.

3.4.2. Farm monitoring plan

All the farms were monitored for 6 months to assess the status of BMSSC, record the incidence of CM and dynamic epidemiological information at farm and animal levels. A functional communication network was developed between CVASU mastitis research team and the farmers to execute farm monitoring. At the beginning of the study, an interactive training programme was organized for the farmers to understand and learn how to operate the network in recognizing CM cases, collecting samples and recording required information. The questionnaire was thoroughly discussed to record necessary data and a set of clinical signs of mastitis with images were provided to each farmer for easy recognition of different categories of CM in Bengali (local language). After training, the farmers were provided sterilized falcon tubes, cotton, antiseptics, marker pen, tape, simple sample collection procedure flow chart in Bengali language, field visit schedule for dynamic data collection and basic case information recording sheets. The questionnaire, case recording sheet and mastitis categorization sheet have been attached in the **Appendix-II**.

3.4.3. Dynamic data collection

For the farm dynamic data, each selected farm was visited at one-month interval during the study period. The dynamic data recorded were: i) number of heifers turned into lactation, ii) number of lactating cows sold, iii) number of cows dried, iv) number of lactating cows died, v) any other loss, vi) any changes in farm management (feeding, cleaning, milking technique etc.), vii) climatic conditions (temperature and humidity). A digital hygrometer (PRESTO® digital hygrometer, model: 110, origin: Taiwan) was used to measure the temperature and humidity at each month interval. Animal level data (age, parity, BCS, milk yield and days after parturition) for each

newly recruited lactating cow (newly calved/bought) were collected. During the first and subsequent farm visits farmers were asked about management of CM therapeutically and non-therapeutically (if experienced mastitis) and the information were noted.

3.4.4. Definition and categorization of clinical mastitis

Clinical mastitis was classified according to criteria described by Pinzon-Sanchez and Ruegg, (2011) : i) Grade-1 (mild), when only the milk was abnormal; ii) Grade-2 (moderate), when abnormal milk was accompanied by swelling and/or redness and/or pain of mammary gland; iii) Grade-3 (severe), when the cow exhibited systemic signs of illness such as depression, anorexia, dehydration, or fever in addition to milk and udder changes.

Whenever any farmer identified a cow with CM, the farmer was requested to record observable clinical signs in a pre-structured record keeping sheet and to collect and store milk samples temporally in the refrigerator before giving any drug. The farmer of the affected cow's farm was requested to inform the CVASU mastitis team immediately to visit the farm for collecting samples and check and record required information at individual animal (such as BCS, lactation stage etc.) and quarter level (such as size, shape, distance between tip of teat and floor).

3.5. Sample collection, preservation and transportation

Clinical mastitis milk samples were collected from every new case.

3.5.1. Quarter milk samples for bacteriology

Milk samples, 5 mL from each infected quarter, were collected from cows with CM, aseptically by assigned trained farmers in the selected farms using a 15mL sterilized falcon tube supplied prior to each visit. The detailed sample collection procedure is given in **Appendix-III.** The milk samples were transferred within 3 hours to the laboratory of CVASU in insulated ice-box and stored in -20°C before the samples were cultured for pathogen isolation and identification.

3.5.2. Sample collection for bulk milk somatic cell count

Bulk milk samples were collected from all studied farms at two-month interval for SCC using a somatic cell counter (DeLaval Group, Stockholm, Sweden; Sensitivity: 88% and Specificity: 80%). The collected bulk milk (25-50 mL) contained in a 50 mL falcon tube, was transferred to the laboratory in insulated ice-box and SCC was

performed immediately after arrival at the laboratory of Department of Medicine and Surgery, CVASU. The detailed procedure of SCC is given in **Appendix III**.

3.6. Isolation and identification of bacteria

A total of 155 samples were collected from 204 affected cows (N=1383 cows) with CM were undertaken for bacteriological evaluation in accordance with NMC protocol. *S. aureus*, NAS, *Str* spp., *Bacillus* spp., *Escherichia coli*, *Klebsiella* spp. *Enterobacter* spp. and *Pseudomonas* spp. were considered for isolation from the collected CM milk samples. Isolates were preserved at -80°C using Brain Heart Infusion Broth and 50% Buffered glycerol for future use. The detailed procedure has been attached in **Appendix-IV**.

3.7. Matrix Assisted Laser Desorption/Ionization Time of Flight

Fourteen isolates of *Staphylococcus* spp. including the *S. aureus* isolates were transferred to the reference mastitis laboratory, SVA, Sweden. After re-culturing into blood agar, the pure colonies were used for Matrix assisted laser desorption/ionization time of flight (MALDI-TOF) confirmation. Part of single colony were touched with sterile toothpick and smeared correctly in sample plates for MALDI-TOF. Single sample was smeared in two separate wells for repeated testing of the isolate. Then matrix solution is added over each well in sample plate and inserted into MALDI-TOF device (Bruker rapifleX®, Bruker Daltonik GmbH, Bremen, Germany). With higher spectra best matched organism was detected matching with micro-organisms in database.

3.8. Minimum inhibitory concentration determination

Two confirmed isolates of *S. aureus* were subjected to MIC determination in micro dilution method using antibiotics coated microtitre plates (VetMIC CLIN staf/strept panel (Ver. 2016-05; Lot no. 168/16; Art no. E395129, Statens Veterinarmedicinska Anstalt, Uppsala, Sweden). To validate the test, *S. aureus* CCUG 15915 which is an analogue to *S. aureus* ATCC 29213 was used as a control.

3.9. Statistical analysis

Data obtained, were entered into Microsoft Excel 2016. Data cleaning, coding and integrity checking were performed in the Excel programme before exporting to STATA IC-13 (StataCorp, 4905, Lakeway Drive, College Station, Texas 77845,

USA) for epidemiological analysis. R studio 1.2.1335 (Free Software Foundation, Inc. 51 Franklin Street, Fifth Floor, Boston, MA 02110-1301, USA) was used to design some graphs.

3.9.1. Descriptive statistics

At first, the average value of BMSCC (cells per µL) of 3 consecutive measurements (with a 2-month interval) per farm was calculated before conducting summary statistics on average values of BMSCC of all studied farms. The results were presented as means with quartile distribution.

Incidence rate of CM at animal level was calculated as the number of animals with the disease out of the total number of animal-time at risk. The IRCM at quarter level was calculated as the number of quarters with the disease out of the total number of quarter-time at risk. The animal level and quarter level IRCM within each farm was computed as described earlier for calculation of animal level incidence rate. The IRCM at animal and quarter level were also calculated separately for each grade of mastitis as described before. The results were expressed as cases per 100 cows or quarters per year.

Proportionate prevalence was estimated by dividing the number of samples positive to each of the tested pathogens (*Str* spp./ NAS/ *S. aureus/ Bacillus* spp./ *E. coli/ Pseudomonas* spp./ *Enterobacter* spp./ *Klebsiella* spp.) by total number of samples tested. The prevalence of the pathogens was as distributed by farm, animal and quarter levels. Frequency distribution of farmers' attitude toward mastitis and mastitis treatment were calculated as number of responses against each query divided by the total number of respondents. The results were expressed as frequency numbers and percentages.

3.9.2. Risk factors analysis

Risk factors were analyzed for mean BMSCC with farm level factors and for CM with individual animal level factors.

3.9.2.1. Univariable analysis

Farm level factors were recorded into two or three categories based on meaningful number of frequency. Either t-test or 1-way ANOVA was carried out to test the association between BMSCC and each of 19 selected farm level factors. These factors

included farm size, year of establishment, number of employees, age of farmer, educational qualification, number of milking cows, housing system, source of animals, water refreshed per day, footbath, floor cleanliness, farm environment cleaning frequency, floor cleaning frequency per day, draining score, milking system, calf sucking, number of milkers, environmental temperature and humidity. Farm size, number of employees, age of farmers, number of milking cows, number of milkers, temperature and humidity; these quantitative factors were categorized into two or three based on their percentiles for ensuring optimum frequencies under each category of respective factors. Quantitative factors were not used as their distribution was skewed even after log transformation. Significant Factors at $p \leq 0.2$ level of significance were used for developing multivariable regression model.

Chi-squared tests were conducted to evaluate the association between the binary response variable of CM in cows and each of 19 selected factors. Quantitative factors were converted into categorizes based on their percentiles. Categorical factors were regrouped into few groups to ensure enough observations in each category. Body condition score was categorized into three classes: poor (≤3), moderate (3.1- to 3.25) and fat (3.26 to 4); parity into five:1, 2, 3, 4 and 5 or more and days after parturition was divided into four classes: 27 or less, 28 to 90, 91 to 185 and 186 to 480 or more.

3.9.2.2. Linear model

The model was manually constructed by backward selection applying the maximum likelihood estimation procedure and the statistical significance of the contribution of individual factors (or group of factors) was determined using Wald's test and the likelihood ratio test (LRT) (Dohoo et al., 2003).We assessed for interaction between factors by constructing two-way interaction product terms for the significant main effect factors in the model, forcing them into the model and examining changes in the coefficients and *p* values of the main effects. The presence of confounding factors was investigated by removing one of the variables and assessing changes in the beta-coefficients of the remaining variables. A coefficient change of more than 10% was considered to indicate confounding. Variance inflation factors (VIF) for the factors were examined to diagnose collinearity and to identify highly correlated factors to avoid a duplication of effects. A VIF value of more than 10 indicates serious collinearity (Dohoo et al., 2003). The Cook–Weisberg test was used to examine the

homogeneity of variance and whether the overall data fitted the model. The results were presented for each adjusted factor as a beta-coefficient, p value and 95% CI.

3.9.2.3. Univariable random effects logistic regression

Factors significant at $p \leq 0.2$ in the chi-squared test were tested in a univariable random effects logistic regression model taking Farm ID as a random effect to evaluate the strength of association between the binary response variable CM (Yes or No) and the explanatory variables. Body condition score, lactation stage and milk yield were included in this model. The analysis was carried out as described by logistic regression model with random effects (Dohoo et al., 2003). The results were presented as odds ratio (OR), *p* value and 95% CI. The results of the univariable random effects logistic regression models were presented as subject specific odds ratio's to show the effect of the risk factors within farms rather than across farms. Multivariable logistic regression model construction was not possible due to significant correlation among the factors (Milk yield, BCS and lactation stage) retrieved from univariable analysis.

Chapter-4: Results

4.1. Herd characteristics

Farmers had a diverse educational qualification from Secondary or less (33%) to graduation or higher than that (67%). Farms were maintained by employees varied from 3 to 58 in number. Selected farms had cattle population ranging from 26 to 353 of which lactating cows varied from 9 to 235. In case of 79.2% farms, farmers allowed entry of new animals from both of its own stock and also from other sources. Farm environments (shed surroundings) of 79.2% farms were cleaned 1-4 times per day, whereas farmers of 20.8% cleaned in a weekly basis or even after long period. A water refreshing schedule of once or twice per day was maintained by 54.2% farms. There were no disinfectant foot bath in 87.5% farms at the entry point as a regular hygienic practice. Farmers cleaned their shed floor 4 to 6 times a day in 75% farms. Though 54% farms had either excellent or very good drainage system, around 88% farms' floor were wet on observation. Majority of the farms (75%) practiced hand milking system where at least 50% farms had 1 to 4 milkers. In 50% farms, calf suckling was strictly avoided while in 20.8% farms farmers allowed calf sucking and in 29.2% farms both were maintained based on individual animal. Average temperature and humidity within farms were in between 27.2°C to 39.8°C and 62% to 82.2%, respectively.

4.2. Bulk milk somatic cell count

Among 24 farms, the lowest mean of BMSCC of any farm was found 354,330 cells per mL whereas highest mean was 1,156,670 cells per mL of milk. Quartile estimates of the average BMSCC of the 24 farms in this study has been described below (Table 4.1).

Table 4.1. Mean bulk milk somatic cell count obtained from 24 farms

Unit of measurement	Somatic cell count (cells per mL of milk)
1st quartile	354,330-597,000
2nd quartile	597,001-777,500
3rd quartile	777,501-932,330
4th quartile	932,331-1,156,670

4.3. Incidence rate of clinical mastitis at different level

At animal level, overall IRCM was 43.9 cases per 100 cows per year. The IRCM of grade-1 was 10.5 cases per 100 cows per year (95% CI: 7.9-14.0), for grade-2 it was 25.5 cases per 100 cows per year and for grade-3 the IRCM was 7.8 cases per 100 cows (95% CI: 5.6-10.9) per year (Table 4.2).

Table 4.2. Animal level incidence rate of clinical mastitis in commercial dairy farms in Chattogram, Bangladesh

Level	Status	Incidence rate (cases per 100 cows per year)	95% CI
Animal level	Overall	43.9	38.2 to 50.5
	Grade-1	10.5	7.9 to14.0
	Grade-2	25.5	21.3 to 30.7
	Grade-3	7.8	5.6 to 10.9

95% CI: 95% confidence interval

Overall quarter level IRCM was 11.6 cases per 100 quarters per year. According to grade the IRCM was 6.5 cases per 100 quarters per year (grade-2), which was higher than grade-1 or grade-3 (Table 4.3). According to quarter position, the overall IRCM was more higher in front quarters compared to IRCM in hind quarters while IRCM in front left quarter was much higher than front right and in hind quarters IRCM was more in right quarter than the left one (Table 4.3).

Table 4.3. Quarter level incidence rate of clinical mastitis in commercial dairy farms in Chattogram, Bangladesh

Level	Status	Incidence rate (cases per 100 quarters per year)	95% CI
Quarter level	Overall	11.6	10.2 to13.3
	Grade-1	2.8	2.1 to 3.6
	Grade-2	6.5	5.5 to 7.8
	Grade-3	2.4	1.8 to 3.2
Quarter position	Front right	10.8	8.3 to 14.2
	Front left	14.2	11.1 to 18.0
	Hind right	13.1	10.2 to 16.8
	Hind left	8.5	6.3 to 11.6
Quarter position	Front	12.5	10.4 to 15.0
	Hind	10.8	8.9 to 13.1

95% CI: 95% confidence interval

Between farms, the IRCM ranged from 0 (0-23.2) to 103 (95% CI: 62.4 to171.7) cases per 100 cows in a year (Figure 4.1).Within farm IRCM varied from 0 (95% CI:

0- to 6.5) to 52.7 (95% CI: 27.3 to 92.3) cases per 100 quarters in a year (Figure 4.2). No cases of CM were detected in two of the farms.

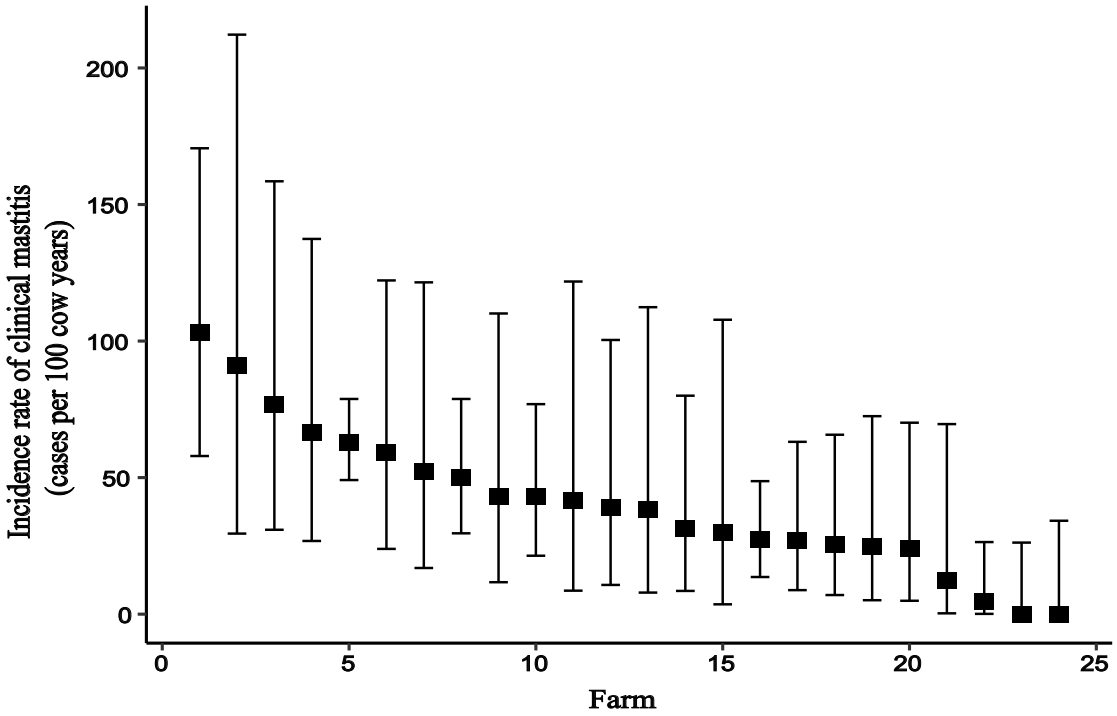

Figure 4.1. Farm level incidence rate (100 cows per year) of clinical mastitis in the dairy farms designated by numbers

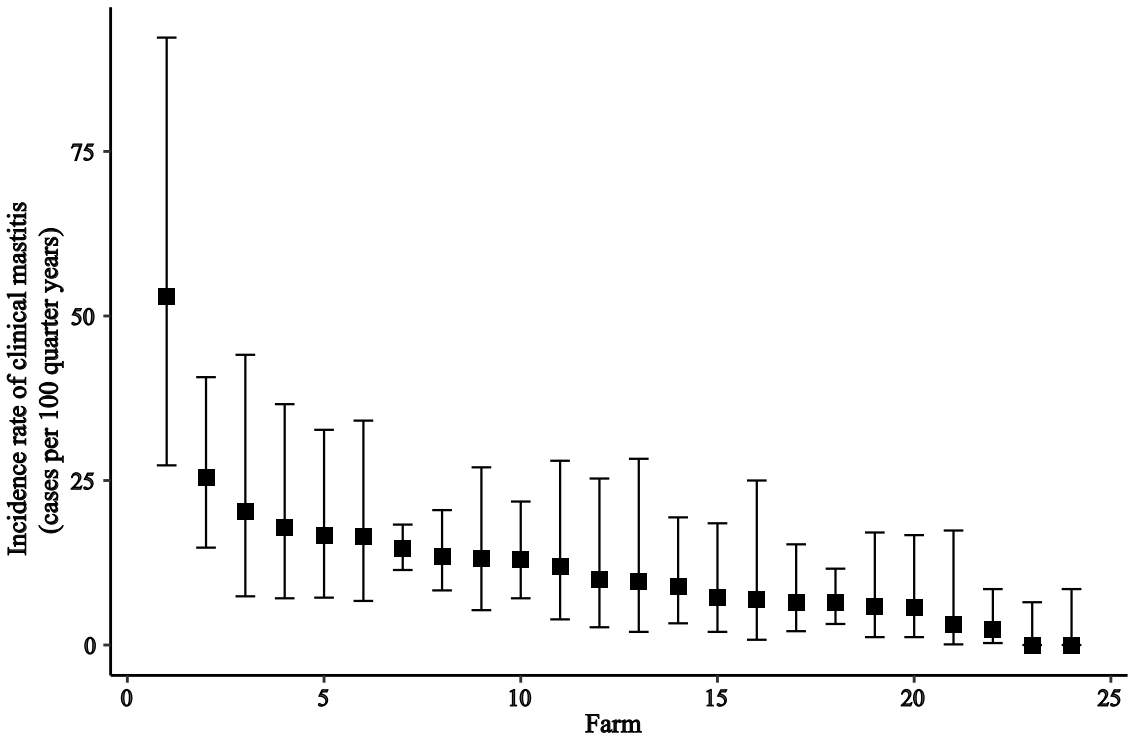

Figure 4.2. Farm level incidence rate (100 quarters per year) of clinical mastitis in the dairy farms designated by numbers

4.4. Risk factor analysis

4.4.1. Univariable association between bulk milk somatic cell count and farm level factors (t-test or 1-ANOVA)

The mean BMSCC was significantly associated with five factors at p≤0.2: i) Source of animals, ii) farm environment cleaning frequency, iii) milking system and iv) humidity (%) (Table 4.4.1).

Table 4.4.1. Univariable associations between the farm level factors and mean bulk milk somatic cell count of lactating cows during the period of 6 months (n=24 farms)

Variable	Categories	n	Mean BMSCC ± SD	*p*
Farm size	26 - 65	13	712.9± 184.6	0.3
	66 - 353	11	847.7± 261.4	
Year of establishment	1978 - 2004	12	724.5± 215.0	0.7
	2004 - 2016	12	824.9± 239.6	
Number of employees	3 - 8	14	762.6± 227.8	0.8
	9 - 58	10	791.5± 240.7	
Age of farmer (years)	24 - 40	14	769.0± 66.9	0.6
	41 - 60	10	782.5± 65.4	
Educational qualifications	Secondary or less	6	856.6± 181.2	0.06
	Higher secondary	2	683.3± 3.8	
	Graduation or higher	16	755.4± 254.7	
Number of milking cows	9 - 24	14	757.3± 228.8	0.9
	25 - 235	10	799.0±238.0	
Housing system	Face-in	7	687.9± 240.5	0.3
	Face-out	10	855.1± 210.3	
	Both	7	746.6± 235.3	
Source of animals	Both	19	778.8± 219.6	0.1
	Own stock	4	659.6± 210.5	
	Purchase	1	1156.7	
Water refreshed per day	1-2	13	806.7± 243.3	0.8
	3-6	9	739.4± 220.5	
	Auto drinker	2	725.5± 268.5	
Footbath	Yes	3	803.8± 260.4	0.6
	No	21	770.5± 230.4	
Floor cleanliness	Dry	3	561.2± 220.8	0.8
	Wet	21	805.2± 217.6	
Farm environment cleaning frequency	1-4 times per day	19	808.9± 240.4	0.2
	Weekly or more	5	644.7± 115.7	
Floor cleaning frequency per day	2 - 3 times	6	787.6± 241.1	0.8
	4 - 6 times	18	770.4± 231.2	
Draining score	Excellent and very good	13	754.4± 250.2	0.6
	Good and poor	11	798.6± 209.1	
Milking system	Manual	18	781.6± 60.4	0.1
	Manual and Machine both	6	753.8± 52.4	
Calf sucking	Yes	5	795.1 ± 253.7	0.9

	No	12	784.3 ± 248.4	
	Both	7	743.5 ± 205.8	
Number of milkers'	1 - 3	12	737.6± 215.0	0.7
	4 - 16	12	811.8± 244.8	
Within farm temperature	27.2 - 30	12	809.2± 225.6	0.9
(°C)	30.1 - 39.8	12	740.2± 235.9	
Within farm humidity (%)	62 - 75	11	813.1± 290.1	0.07
	75.1 - 82.2	13	742.1± 165.5	

n: Frequency number; Mean BMSCC: Mean of bulk milk somatic cell count; SD: Standard deviation

4.4.2. Multivariable linear regression model for bulk milk somatic cell count

Neither confounding nor interaction was detected during the model building process. Variance inflation factors remained below 10, indicating the factors were not collinear. The Cook–Weisberg test p-value of 0.9 suggested the model fitted well. According to the final model, own stock as replacement (p=0.09), and 1 to 4 times or more cleaning (p=0.1) were associated with higher level of BMSCC (Table 4.4.2).

Table 4.4.2. Outputs of multivariable linear regression between the farm level factors and mean bulk milk somatic cell count of lactating cows during the period of 6 months (n=24 farms)

Variables	Categories	β	95% CI	*p*
Source of animals	Both	Ref		
	Purchase	295.1	-130.9 to 721.1	0.2
	Own stock	-202.0	-437.4 to 33.4	0.09
Farm environment cleaning frequency	1-4 per day	Ref		
	Weekly or more	-168.0	-382.2 to 46.1	0.1

β: Correlation co-efficient; 95% CI: 95% confidence interval; p: probability

4.5. Associations of the incidence rate of clinical mastitis with potential risk factors

Tables 4.5 gives the descriptive information and univariable associations between the IRCM and five cow-level potential risk factors: BCS, number of parity, daily milk yield, lactation stage and days after parturition. A positive association was observed with CM and BCS and a very strong association with high milk yield. Early lactation cows were at higher risk of CM than cows later in their lactation.

Table 4.5. Univariate association and incidence rate of clinical mastitis in dairy cows by different factors

Factors	Categories	N	Rate per 100 cows per year	95% CI	Odd ratio	95% CI (OR)	*p*
Body condition score	3 or less	427	34.9	26.5-45.9	Ref		
	3.1-3.25	488	45.7	36.0-58.0	1.3	0.9-1.9	0.2
	3.26 or more	468	51.0	40.8-63.7	1.5	1.0-2.2	0.04
Parity	1	364	42.1	31.9-55.6	Ref		
	2	354	42.9	32.4-56.8	1.1	0.7-1.7	0.7
	3	279	37.5	26.9-52.2	1.1	0.7-1.7	0.7
	4	214	51.8	37.2-72.1	1.3	0.8-2.0	0.3
	5 or more	172	51.4	35.3-75.0	1.1	0.7-1.9	0.7
Milk yield (Liter)	10 or less	462	30.8	23.0-41.3	Ref		
	10.1-13	266	38.6	27.6-54.0	1.4	0.9-2.3	0.1
	13.1-17	328	38.8	28.8-52.4	1.4	0.9-2.2	0.1
	17.1or more	327	73.0	58.1-91.7	2.8	1.9-4.1	<0.001
Lactation stage	Early	748	55.0	46.0-65.7	1.6	1.1-2.4	0.01
	Mid	283	33.8	25.2-48.0	1.2	0.8-2.0	0.4
	Late	352	31.7	23.1-43.6	Ref		
Days after parturition	27 or less	354	51.4	38.1-69.3	1.4	0.9-2.2	0.2
	28-90	392	56.2	44.9-70.4	2.1	1.3-3.2	0.001
	91-185	292	38.3	28.3-51.8	1.5	0.9-2.4	0.1
	186 or more	345	29.7	21.3-41.4	Ref		

95% CI: 95% confidence interval; OR: Odd ratios; p: Probability

4.6. Bacteriological culture results of milk samples from clinical mastitis

Of 153 quarter milk samples from CM cases, 16.2% samples were found negative during BC and 1.8% samples were contaminated. The other samples were found positive where 167 bacterial isolates were identified. NAS was the most frequently isolated organism (24.0%) followed by *Bacillus* spp. (19.8%), *Str*. spp. (18.6%), *E. coli* (6.0%), *Klebsiella* spp. (4.2%) and *Pseudomonas* spp. (4.2%) (Table 4.6).

Table 4.6. Pathogen distribution (n=167) from 153 clinical mastitis affected quarter samples in 24 farms

Name of pathogens	N=167	%	95% CI
Str. spp.	31	18.6	12.6 to 26.3
NAS	40	24.0	17.7 to 31.2
S. aureus	6	3.6	1.3 to 7.7
Bacillus spp.	33	19.8	13.6 to 27.8
E. coli	10	6.0	2.9 to 11.0
Pseudomonas spp.	7	4.2	1.9 to 8.6
Enterobacter spp.	3	1.8	0.4 to 5.2
Klebsiella spp.	7	4.2	1.9 to 8.6

No growth	27	16.2	10.7 to 23.5
Contaminated	3	1.8	0.4 to 5.2

%: Percentages; 95% CI: 95% confidence interval

4.7. Matrix Assisted Laser Desorption/Ionization Time of Flight testing of selected isolates

By using BC, 167 bacterial isolates were evaluated and 14 isolates of *Staphylococcus* spp. were selected for MALDI-TOF. Among these 14 isolates, 2 isolates were confirmed as *S. aureus* and 12 were identified as NAS (Table 4.7).

Table 4.7. Matrix Assisted Laser Desorption/Ionization Time of Flight results of the examined isolates (N=14)

Bacteria	Number (%)
Staphylococcus sciuri	5 (35.7)
Staphylococcus haemolyticus	4 (28.6)
Staphylococcus aureus	2 (14.3)
Staphylococcus arlettae	2 (14.3)
Staphylococcus cohnii	1 (7.1)

%: Percentages

4.8. Minimum inhibitory concentration determination of isolated *S. aureus*

Two *S. aureus* isolates underwent antimicrobial susceptibility testing to determine the MIC. Among the panel of antibiotics, one isolate of *S. aureus* was found sensitive to the antibiotics. Another isolate was Methicillin resistant *S. aureus* (MRSA) which was also confirmed by PCR and found resistant against 5 antibiotics (Penicillin, Oxacillin, Cefoxitin, Gentamycin and Tetracycline) (Table 4.8).

Table 4.8. Antimicrobial susceptibility result of two selected isolates

Antibiotics	**ID of *S. aureus***	**Distribution of MIC**											
µgm per mL		**0.03**	**0.06**	**0.12**	**0.25**	**0.5**	**1**	**2**	**4**	**8**	**16**	**32**	**64**
Penicillin	1						**R**						
	2	**S**											
Cefalotin	1						**S**						
	2						**S**						
Oxacillin+2% Nacl	1						**R**						
	2				**S**								
Cefoxitin	1									**R**			
	2				**S**								
Enrofloxacin	1				**S**								
	2				**S**								
Fusidic acid	1					**S**							
	2					**S**							
Erythromycin	1					**S**							

	2					**S**							
Clindamycin	1					**S**							
	2					**S**							
Gentamycin	1								**R**				
	2						**S**						
Nitrofurantoin	1										**S**		
	2										**S**		
Tetracycline	1								**R**				
	2					**S**							
µgm per mL		**0.25/4.75**	**0.5/9.5**	**1/19**	**2/38**	**4/76**							
Trimethoprim + Sulfonamide	1	**S**											
	2	**S**											

MIC: Minimum Inhibitory concentration; ID: Identification number; R: Resistant; S: Sensitive; Concentration in blocked boxes were absent in the MIC plate

4.9. Farmers perception and behavior and treatment strategies toward clinical mastitis

Farmers opinions on their perception and behavior and treatment strategies for CM were described in Table 4.9. Within all farmers, 30% responded that unhygienic conditions and incorrect milking procedure caused mastitis in their farms. According to the farmers' responses most commonly used antibiotics were gentamycin (20%), followed by ceftriaxone (17%) and amoxicillin (11%). Majority of the farmers (53.6%) said that they use probiotic as a supporting drug in case of mastitis, and 7% do not use any supporting drugs. Non-steroidal anti-inflammatory drugs (NSAID) (7.1%), homeopathic drugs (3.6%), antihistamines (3.6%), dexamethasone (3.6%) and vitamin B complex (3.6%) were also used. For mastitis therapy, 45.8% farmers replied that they initially started treatment of mastitis with drugs (such as antibiotics) before consulting with a veterinarian. Farmers also considered mastitis treatment by traditional ways with phytolocca, a homeopathic drug (32.4%), ice application (11.8%), naphthalene (8.8%) and cold water (5.9%). All of the farmers (100%) answered that CM is problematic and they had culled at least one lactating cow due to poor udder health when they were non-responsive to any kind of treatment. None of the farmers follow withdrawal period of antibiotics in milk following antibiotics treatment and sell those milk and all of them discard the milk from CM affected cows due to curd formation.

Table 4.9. Descriptive analysis of farmers perception towards mastitis and treatment strategies for clinical mastitis

Variables	Categories	Frequency (%)
What do you think about CM? (n=30)	Unhygienic condition causes mastitis	9 (30)
	Milking fault causes mastitis	9 (30)
	Managemental fault causes mastitis	3 (10)
	This is a bacterial infection	3 (10)
	This results in udder swelling	2 (6.7)
	This is a water borne infection	1 (3.3)
	High yielding cows are repeatedly affected	1 (3.3)
	This results in milk color change	1 (3.3)
	Mysterious thing	1 (3.3)
What are the commonly used antibiotics? (n=91)	Gentamycin	20 (22.0)
	Ceftriaxone	17 (18.7)
	Amoxycillin	11 (12.1)
	Colistin	6 (6.6)
	Kanamycin	5 (5.5)
	Neomycin	5 (5.5)
	Ciprofloxacin	4 (4.4)
	Oxy-tetracycline	4 (4.4)
	Streptomycin-Penicillin	4 (4.4)
	Ampicillin	3 (3.3)
	Amoxicillin-Cloxacillin	2 (2.2)
	Gentamycin intra-mammary tube	2 (2.2)
	Penicillin	2 (2.2)
	Sulfamethoxazole-trimethoprim	2 (2.2)
	Ampicillin-Gentamycin	1 (1.1)
	Ceftiofur	1 (1.1)
	Marbofloxacin	1 (1.1)
	Sulfadimidine-Trimethoprim	1 (1.1)
What are the supporting drugs used? (n=28)	Probiotics	15 (53.6)
	No supportive drugs	7 (25.0)
	NSAID	2 (7.1)
	Homeopathic	1 (3.6)
	Antihistamines	1 (3.6)
	Dexamethasone	1 (3.6)
	Vitamin B-complex	1 (3.6)
Who prescribes the drugs? (n=24)	Self then doctor	11 (45.8)
	Self	9 (37.5)
	Veterinarian	3 (12.5)
	Animal husbandry officer	1 (4.2)
What are the traditional treatments used? (n=34)	Phytolocca (Homeopathic drug)	11 (32.4)
	Ice	4 (11.8)
	No traditional treatment	3 (8.8)
	Napthalene	3 (8.8)
	Cold water	2 (5.9)
	Egg yolk emulsion	2 (5.9)
	Belladona (Homeopathic drug)	2 (5.9)
	Calcarea sulf (Homeopathic drug)	2 (5.9)
	Walking	1 (2.9)
	Heat	1 (2.9)
	Turmeric	1 (2.9)
	Coconut oil	1 (2.9)
	Aurum mureacum (Homeopathic drug)	1 (2.9)

%: Percentages; NSAID: Non-steroidal anti-inflammatory drugs

Chapter-5: Discussion

Scientific scrutiny of mastitis in dairy cows in this country is mostly based on cross-sectional studies (Rahman et al., 2010; Islam et al., 2011; Sarker et al., 2013b; Hoque et al., 2015). Cross-sectional study cannot efficiently determine the temporal association of exposure and effects on occurrence of specific outcome. As a result, effective strategies have not previously been identified to prevent and control mastitis. Therefore, the present cohort study (longitudinal) was conducted to fill the previous gaps of mastitis epidemiology with the target of recommending better preventive and control strategies for mastitis. In this section of the thesis important results of the present study, their implications, limitations and conclusion and recommendations have been discussed.

5.1. Herd characteristics

Variable farm sizes, predominantly own source to restock the farm animals, both faces-in and face-out system, mostly wet floor, absence of foot bath (in most of the cases) and high poor drain score, mostly hand milking, access to calf suckling were common features of the studied farms which partly reflect usual scenarios of dairy farming standard across Bangladesh (Rahman et al., 2009; Parvez et al., 2015; Sultana et al., 2015). These scenarios are very favorable for different mastitis pathogens to spread between farms and cows, and infect udders (Kivaria et al., 2007; Oliveira et al., 2015). Therefore, interventions are needed to improve the dairy farm practices to reduce the burden of pathogens and increase production potential ensuring food safety. The present study found the majority of the farmers being educated which suggest that they could adapt new technologies and better farm management practices. These farmers could upgrade their knowledge if they were provided training and simple manuals or leaflets considering the risk of faulty management practices.

5.2. Status of bulk milk somatic cell count

In the present study, 20 of 24 farms had a BMSCC exceeding 400×10^3 cells per mL milk which is the legislative limit in many developed countries (such as: for EU 400,000 cells per mL of milk), (Barkema et al., 1998; Sargeant et al., 1998) suggesting that there might have been more cases of SCM in the herds. The lowest quartile (25%) had BMSCC values of 354×10^3 to 597×10^3cells per mL milk. It is

therefore logical to set the threshold value of BMSCC for other farms in this region to improve and reach at this level by improving farm management practices like pre- and post-milking teat disinfection, teat drying before milking, proper management of milking machine (Vissio et al., 2018). The additional measures for controlling high BMSCC are to reduce the new infection rate (NIR) and decreasing duration of infection. Pre- and post-milking disinfection will help decrease the NIR and culling of chronically infected cows will reduce the duration of infection (Barkema et al., 2013). Moreover, milking precautions like keeping cows standing following milking, wearing gloves during milking and milking the affected cow at last should be implemented for better results (Dufour et al., 2011; Barkema et al., 2013).

5.3. Incidence rate of clinical mastitis at different levels

The IRCM was fairly high and a corncern at quarter level in the present study. Incidence rate of CM at cow and quarter level raised up to 43.3 cases per 100 cows and 11.6 cases per 100 quarters per year respectively. These results are quite alarming in dairy industry in Chattogram, and consistent with the findings of many studies reported in other parts of the world (Bradley et al., 2007; Kivaria et al., 2007). The high IRCM in this study has relevance with inadequate farm mangement practices (Sarker et al., 2013b) and lack of loose housing system in farming situation of Bangladesh (Kalmus et al., 2006). Within farm, IRCM varied from 0 to 103.5 cases per 100 cows per year and at quarter level the range found was 0 to 52.9 cases per 100 quarters in a year which is supported by another study (Olde Riekerink et al., 2008). The large variation of IRCM within farm is likely the result of variation in farm management practices. Therefore, there are good improvement opportunities in farms with very high IRCM.

The IRCM was significantly higher for grade-2 at both animal and quarter levels compared to grade-1 and grade-3 in the present study which is not agreed by Pinzon-Sanchez and Ruegg, (2011) and Verbeke et al. (2014). These studies showed the predominance for grade-1 CM. Reasons behind this variation might be due to fluctuation in diagnostic precision for differentiating the different grades of CM or distributional differences as burden of CM may vary depending on distribution of CM pathogens at various geographical locations. But these findings suggest that a better

farm monitoring system should be implemented to detect grade-1 mastitis to manage overall mastitis burden in farms.

In many studies, IRCM was calculated at cow level but very few studies were undertaken to estimate IRCM on quarter level (Kivaria et al., 2007; Tomazi et al., 2018). However, more specifically IRCM at fore or hind quarters were not displayed in any other studies. But the high incidence of CM in front quarters was not agreed by earlier study (Bhat et al., 2017) which might be due to association of other factors like shape of teat end and previous teat injury which may increase the susceptibility for getting mastitis. But this was not concluded in the findings of the present research.

5.4. Risk factors associated with bulk milk somatic cell count

Source of animals (p=0.09) and floor cleanliness (P=0.06) were borderline significant factors associated BMSCC in the current study.

Animals purchase from other farms increased the level BMSCC which was not in agreement with the report by Tolosa et al. (2015) as purchase of new animals may bring some chronically affected cows. However, it is promising finding that animals restocked with own farm animals was associated with a lower BMSCC level which suggests cows from own farm stock might be more adaptable with the existing farming condition than newly brought animals. So, due to strong herd immunity responses own stock animals are more consistent with preventing infection (Schukken et al., 2011). Allowing lactating cows in herd from other sources rather than own stock might introduce contagious pathogens in lactating cows which resulted high BMSCC.

Dry floor was significantly associated with decreased level of BMSCC. This finding is also consistent with Santman-Berends et al. (2016), who also found that shed with cleaning of floor once daily had more number of CM incidence compared to cleaning 4 times a day. This can be explicated as the presence of moisture is necessary for survival and multiplication of bacteria. Moist bedding materials can easily be adhered with cows and facilitate transfer of organisms from bedding materials to cow skin which in turn will responsible for increased level of BMSCC (Fávero et al., 2015).

5.5. Factors associated with clinical mastitis

Multivariable logistic regression model was not constructed as there were collinearity among BCS, lactation stage and/or milk yield. There were associations between body weight and BCS, and BCS and lactation stage which were also found in another study (Berry et al., 2007). The nature of BCS causing mastitis could vary in response to variation in number of parity and milk yield trait.

Cows with higher BCS (3.26 or more) were more susceptible to CM cases (OR 1.5) which is surprising result, not consistent with many previous studies (Kivaria et al., 2007; Sarker et al., 2013b), which found that poor BCS increased the occurrence of CM (OR 7.3) (Sarker et al., 2013a). Although few studies (Gundelach et al., 2011; Tolosa et al., 2015) agreed with positive correlation of IMI having good BCS. However, the actual association of good BCS was not further explained. Cross-bred cows usually have a good BCS and yield more milk (in Bangladesh situation) which could have linked up with more CM cases in Bangladesh. This explanation can further be expanded by describing correlation of BCS with some other factors in association with CM. Cows with higher BCS are susceptible to some other diseases like sub-clinical ketosis and milk fever which may lead to cows at an immunosuppressive state and that can be associated with CM (Oltenacu and Ekesbo, 1994; Heuer et al., 1999; Gillund et al., 2001; Roche and Berry, 2006; Berry et al., 2007).

The odds of CM of early and mid-lactation stages were 1.6 times and 1.2 times, respectively compared to late lactation. Early lactation was a significant risk factor for CM in this study which is in-lined with the previous studies (Gundelach et al., 2011; Oliveira et al., 2015). It is probably due to the peri-parturient period of two weeks prior and post-calving where the mammary gland remains in a risk for intense growth or differentiation of secretory parenchyma for colostrum accumulation (Oliver and Sordillo, 1988). So, in this period the mammary gland remains in stress condition and may be easy to be invaded by organisms. However, few studies found mid lactation as a risk factor with CM (Mureithi and Njuguna, 2016). This may be explained as in mid lactation stage (3-6 months) the cow remains in peak milk production which is associated with high CM incidence.

In the present study, the cows in early lactation stages but between 28 to 90 days in milk (DIM) had higher IRCM which is also supported by (Kivaria et al., 2007). But

no detailed explanation was found regarding this piece of information. Further longitudinal studies regarding CM incidence needs to be undertaken for precise estimates and finding associations with CM.

High yielding cows were identified as risk factor for CM which is supported by many previous studies (Ericsson Unnerstad et al., 2009; Sarker et al., 2013b; Jamali et al., 2018; Tomazi et al., 2018). Cows yielding large volumes of milk may leak increased level of milk through the teats which will reasonably keep the teat canals open for longer period. This will lead to the risk of infections with mainly environmental mastitis by *E. coli* (Schukken et al., 1991). Cows having very high genetic potentials like milk yield (exotic breeds like Holstein Friesian) was previously reported more susceptible to CM (Hansen et al., 2002). This finding is also logical because the prevalent climatic condition of Bangladesh (high temperature and humidity), and intensive farming system favor the multiplication of organisms in the farms and chances of infections is usually high. High temperature and humidity usually put cross-bred cows on stress and thus compromise body immunity leading to more CM.

But findings on above mentioned risk factors associated with CM should be considered cautiously as they were determined through unadjusted univariable analysis. There is a possibility of existence of confounding bias. Univariate analysis explains association of exposure effects on outcome. But there may be also several extraneous factors which could affect the particular exposure effect on outcome. This could be minimized either by restricting samples or by matching one or more variables.

5.6. Pathogens associated with clinical mastitis

Non-*aureus staphylococci* were the most prevalent organisms in the current study which is consistent with previous studies (Breen et al., 2009; Oliveira et al., 2015; Cheng et al., 2019), and are quite understandable because NAS are very common on udder skins (Spencer and Lasmanis, 1952; Davidson, 1961) and mammary gland tissue (Schalm et al., 1971). Although NAS were previously reported as most common pathogens associated with SCM cases (Pitkälä et al., 2004; Persson et al., 2011), now these had also been identified in many CM cases.

The prevalence of *S. aureus* was low (3.6%), but important because of its zoonotic potential and antimicrobial resistance (Brody et al., 2008). This finding is well

supported by other studies (Waage et al., 1999; Sumathi et al., 2008; Gao et al., 2017; Ruegg, 2018). This can be further explained from a recent detailed molecular study on pathogenesis pattern of isolated *S. aureus* from CM and SCM cases. *S. aureus* produces toxin which can kill neutrophil and hence suppresses immune system of the affected cows (Hoekstra et al., 2018).

The proportion of *Bacillus* spp. was reasonably high (19.9%) which exceeds the proportion reported in other countries (Breen et al., 2009; Kurjogi and Kaliwal, 2011). *Baccillus* spp. are environmental pathogens and may be present when the floor is not properly cleaned and dung and mud is mixed with straw and prevail for longer duration on floor surface. Among the studied farms, some farms used to clean the farm primises less than once in a week which might have played significant role in contaminating the farm environment with the environmental mastitis causing pathogens (Santman-Berends et al., 2016). *Bacillus* may also have been the result of contamination of the samples with straw or bedding materials or the culture media, so may not reflect actual IMI.

Streptococci were one of the highest prevalent pathogen genera (18.6%) obtained from CM cases in current conducted study which was a little higher compared to studies conducted in other parts of the world (Oliveira et al., 2015; Gao et al., 2017; Cheng et al., 2019). This could be explained by the fact that presence of *Streptococci* in the teats and skins of the cows and persists commonly on teat sores. Although various species of *Str*. vary in transmission nature and may be either contageous or environmental as well. *Str. uberis* behaving really like an environmental and *Str. agalactiae* being contagious (generally) and *Str. dysgalactiae* in between. Transmission is mostly occurred through milkers hand and with contaminated floor, utensils and possibly bedding (Constable et al., 2017).

The proportion of *E. coli*, *Pseudomonas* spp., *Entrobacter* spp. and *Klebsiella* spp. was also low (1.8-4.2%) which is in line with multiple published studies nationally and internationally (Sumathi et al., 2008; Verbeke et al., 2014; Gao et al., 2017; Tomazi et al., 2018). These coliform bacteria mostly occupy in cow's environment if not cleaned properly and can easily cause udder infection (Hogan and Smith, 2003).

Microorganisms like NAS and *Str.* were closely associated with either high BMSCC or SCM in most of the previous studies but these were identified as one of the most important CM pathogens in the current study.

5.7. *S. aureus* resistance

The isolate of *S. aureus* that was confirmed methicillin resistant *S. aureus* (MRSA) showed resistance against multiple antibiotics. However, the other *S. aureus* was sensitive to a wide range of antibiotics. The results of presence of MRSA and multi-drug resistance were previously agreed by studies done in other part of world (Vanderhaeghen et al., 2010; Unnerstad et al., 2018). But the present study is not precise enough to compare because only one isolate was MRSA and two isolates were tested for antimicrobial susceptibility testing. Prevalence of MRSA could vary 0-7.5% at herd level (Vanderhaeghen et al., 2010). The transmission can occur between farmers and cows in the herd by direct contact during handling (Unnerstad et al., 2018). The MRSA has been isolated from skin, nose and perineum of animals and other healthy and CM affected cows within the herd may also act as carrier for transmission of MRSA (Unnerstad et al., 2018).

5.8. Farmers perception and behavior and treatment strategies toward clinical mastitis

Most farmers thought, unhygienic condition and faulty milking were the main causes to CM which is also agreed by Sarker et al. (2013b). Farmers were aware about many of the risks and seemed to be motivated to implement mastitis control programme (such as udder cleaning, improvement of stall hygiene, improved feeding and adoptation of fore stripping) which was also suggested by findings of a study conducted in Ethiopia (Mekonnen et al., 2017). As per farmers' response Gentamycin, Ceftriaxone, Amoxycillin and Colistin sulfate were common antibiotics to manage CM therapeutically which suggest the need of establishing treatment guidelines of CM in Bangladesh. Though colistin sulfate is considered as a reserved drug, there are many combined preparations of antibiotics with colistin sulfate are frequently being used by Bangladeshi dairy farmers. Farmers responded that they used Phytolocca, a homeopathic drug (32.4%) and applied ice (11.8%) and egg yolk emulsion (5.9%) to treat CM. Most of the farmers treat their own farm animals by themselves and then consult with a veterinarian (45.8%). This is not an ideal practice to manage CM. This

could be due to lack of required veterinary support in remote areas and availablility of a wide range of antibiotics. Homeopathic drugs are cheaper and have less side effects. As the farmers in Chattogram are educated, they are aware of growing antimicrobial ressistance (AMR) and they may become influenced by success of homeopathic and other traditional treatment used by many farmers. None of the farmers maintain the withdrawal period following antimicrobial therapy that is alarming regarding to public health safety. Clinical mastitis seems to be problematic to any of the farmers as all of them had lost at least one productive cow due to poor udder health. But detailed studies including farmers perception regarding CM has not been undertaken in Bangladesh.

5.9. Limitations of the study

1. Diagnostic error: There might be some contamination; possibly due to sample collection that was done by farmers though they had been trained and unintentional missclassification may be occurred during grading of CM. Bacteriological culture were used for isolation of bacteria from all the samples and only few isolates were retested using MALDI-TOF. The BMSCC testing has a sensitivity and specificity of 88% and 80%. So, this might also intend some minor false positive errors.
2. Statistical assumption and selection bias: Only 24 farms were selected based on easy access and motivation of the farmers but no statistical assumptions was followed for estimating number of farms required. Selection may have caused a bias due to taking convenience sample, probably selecting more motivated farmers and therefore may have possiblity to underestimate or overestimating the true IRCM.
3. Loss of follow-up: No loss of follow up was subjected in the present conducted study. Although loss of follow up is common phenomena for cohort study, this was possible due to small number of farms considered and study period was reasonably shorter.
4. Confounding bias: There may be some more extraneous factors at various levels (farm, cow and quarter levels) which may confound the significant association of particular factors with occurrence of CM.

Chapter-6: Conclusion, Recommendations and Future Directions

6.1. Conclusions

In this study, commercial farming enterprise was initiated by a very diverse group of professional farm owners. Most farmers had education level of graduation or higher studies. They can understand the scientific gaps and be motivated to improve their existing farming management.

The overall farming management in commercial dairy farms in Chattogram was not quite satisfactory. Somatic cell count varied from moderate to very high among the farms. Source of animals in farm population, floor cleaning frequency and farm environment cleaning frequency were found correlated with higher somatic cell count in bulk milk. There is strong possibility that introducing purchased animals from other sources may carry some new infections to the herd. The farmers can improve this situation by creating their own replacement stocks.

Overall IRCM at animal and quarter level were considerably high. Within farm level IRCM at animal and quarter levels varied from very low to very high. The IRCM was higher in front quarters compared to hind quarters. There is room for improvement on most farms with respect to IRCM.

Higher BCS, parity, milk yield and in early lactation stage were more associated with incidence of CM. So, genetic improvement was not proved fruitful in term of prevention question for CM in Bangladesh. High yielding cows were more frequently reported for CM cases. Further assessment is necessary to understand these factors.

Distribution of pathogens from CM milk samples showed that most common pathogen was NAS followed by *Bacillus* spp., *Str.* spp. MRSA was confirmed using PCR and the MIC value confirmed resistance against penicillin, oxacillin, cefoxitin, gentamycin and tetracycline. Regular screening for causal agents and antibacterial susceptibility are necessary to upgrade milk quality.

6.2. Recommendations

1. Somatic cell count estimates for bulk milk varied from low to very high among farms. Farmers should be strict during importation of cows from outside and

animals purchased from outside should be monitored for 3-4 weeks. There should be guideline from DLS for SCC monitoring.

2. High yielding cows suffer from heat stress and they are more possibilities of having CM. In summer there should be options of fogging, so the cows will not suffer heat stress. From own stock, relatively resistant cows to mastitis should be selected as future stock and cows frequently suffer from mastitis should be culled.
3. Within farm IRCM is very high both at cow and quarter levels. There are wide variations between farm to farm IRCM and in few farms, there were no CM cases during the study period. Affected farms should adopt some managemental modifications from better farms like to be cautious in introducing cows in the herd, floor management and strictly adaptation of foot bath.
4. One farm had one case of MRSA. So, affected cows should be regularly identified from new cases and the affected cows should separately be treated or culled for prevention of transmission. Farmers should also be careful about prudent use of antibiotics which might lead to increased level of resistance. Farm personnel should be advocated to use gloves during milking as MRSA has potential zoonotic significance.
5. Selection of antibiotics for treatments should be justified based on antimicrobial susceptibility testing results of disk diffusion or MIC of antibiotics against each specific pathogen depending on the real situations.
6. Farmers should be motivated through workshops, distribution of scientific leaflets or posters to follow judicial use of antibiotics and they should follow the ethics not to sell/mix milk containing antibiotics. There should be a system of monitoring to follow up the improvement.

6.3. Future directions

1. A longitudinal study should be conducted considering a large number of farms with several repeated measurements of BMSCC and a wide range of farm level factors need to be included. This will allow estimating the current status of BMSCC in Bangladesh and specifying the association between wide ranges of farm level factors with high BMSCC.
2. Similar longitudinal and nested case control study could be conducted to other dairy concentrated geographical location of Bangladesh to verify the variation in

countrywide IRCM and association with wide range of farm, animal and quarter level risk factors.

3. Further studies should be conducted to confirm identification of isolated organisms at species level together with molecular testing and their antibiogram.

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Appendix-I

Questionnaire on Investigation of Clinical Mastitis in Dairy Cattle in Chattogram

A. Baseline Information

1. Date of recording/interview [**DATE**]_____/_____/_____

2. Farm identity number [**FID**]_____/_____/_____/_____

3. Name of the farm [**NF**]:_____

4. Type of the dairy farm [**TDF**]

1=Small (_____); 2=Medium (_____); 3=Large (_____)

5. Village [**VIL**]:_____

6.Union [**UN**]:_____

7. Ward [**WD**]:_____

8. Upazilla [**UP**]:_____

9. Longitude [**LON**] and Latitude [**LAT**] of the farm location:_____;_____

10. Year of establishment of the farm [**YEF**]_____

11. Number of farm employees [**EMPLOY**]_____

12. Farmer's demograpghy

12.1. *Name of the farmer [**NF**]*_____

12.2. *Age of the farmer [**AF**]*_____

12.3. *Sex of the farmer [F**SEX**]*_____

12.4. *Education of the farmer [**EF**]*_____

12.5. *Religion of the farmer [**RF**]*_____

12.6. *No of family members [**NFM**]*_____

13. Interviewee's demograpghy (If not the farmer)

13.1. *Name of the farm manger or farm vet [**NM or FV**]*_____

13.2. *Age of the farm manger or farm vet [**AFM or AFV**]*_____

13.3. *Sex of the farm manger or farm vet [FM**SEX or FVSEX**]*_____

13.4. *Education of the farm manger or farm vet [**EFM or EFV**]*_____

13.5. *Religion of the farm manger or farm vet [**RFM or RFV**]*_____

14.Farm's composition

	Types	Total numbers	Breed	Blood percentage
14.1.	*Milking cow[**MC**]*			
14.2.	*Pregnant cow [**PC**]*			
14.3.	*Pregnant heifer [**PH**]*			
14.4.	*Dry cow [**DC**]*			
14.5.	*Non pregnant heifer [**NPH**]*			
14.6.	*Heifer bull [**HB**]*			
14.7.	*Calf [**CALF**]*			
14.8.	*Total [**TOTAL**]*			

15. Housing system of the farm [**HOUSE**]: 1=Face-in; 2=Face-out 3=Both

16. Source of the animal [**SA**]: 1=Own stock; 2=Purchase 3=Both

17. Source of water [**SW**]: 1=Tube well; 2=Deep tubewell; 3=Any other (Pl mention_____)

17.1. *How often water is refreshed? [**WREF**] 1=Three times/day; 2=Four times/day; 3=Any other schedule (Pl mention_____)*

17.2. *Average amount of water provided per lactating cow per day: 1st stage of lactation_____; 2nd stage of lactation_____; 3rd stage of lactation_____*

18. Does the farm have foot-bath at the entrance? [**FOOTB**] 1=Yes; 2=No

19. Cleanliness of farm environment (Gutter, floor etc.)? **[CLENV]** 1=Watery; 2=Dry

19.1. *How often farm environment is cleaned? [**HOCLEAN**]_____*

20. Does the floor of farm suitable for cleaning and disinfecting? [**CLDIS**] 1=Yes; 2=No

20.1. If yes, frequency of cleaning [***FCLEAN***]----------

21. Does the farm have isloation/quarantine shed? [**ISO/QUA**] 1=Yes; 2=No

22. Overall drainage system of the farm [**DRAIN**]:

1=Excellent; 2=Very good; 3=Good; 4=Poor; 5=Very poor

23. Does the farm have boundary wall? [**BOUND**] 1=Yes; 2=No

24. Does the farm have any pest control measure in place? [**PCONT**] 1=Yes; 2=No

25. Feeding system **[FSYST**]. 1=Stall-feeding; 2=Open-feeding; 3=Both

26. Does the farmer have pasture land? [**PAST**] 1=Yes; 2=No

27. Farm milking system [**MILKSY**]: 1=Manual; 2=Machine; 3=Both

27.1. *If manual, sucked by calf [**SUCK**]: 1=Yes; 2=No*

27.2. *If manual, types of milker [TMILKER] 1=New; 2=Old*

28. No of milkers [**MILKER**]:_____

29. Description of milking cows in the farm

No	Tag No	BCS	Age	Parity	Stage of lactation	Milk yield	Date of parturition	Date of entry into the study	Days of milking at entry into the study	Date of exit/event into the study	Days of milking at exit/event into the study

30. Climatic parametrs

31.1.*Ambient temperature [**ATEM**]_____ °F*

31.2. *Humidity [**HUM**]_____*

31.3. *Rainfall [**RAIN**]_____*

31.4. *Radiation [**RAD**]_____*

31. Farmer's behaviour

31.1. *What do you do about mastitis?*

*[**DOMAST**]_____*

31.2. *How often do you check the milk? [**CHMILK**]_____*

31.3.*Do you cull animal because of poor udder health? [**CULL**]* *1=Yes; 2=No*

31.3.1. If yes, when do you exactly cull?[TCULL]----------

31.4. *Do you think clinical mastitis is problematic? [**CMP**]* *1=Yes; 2=No*

31.5 .*What do you do with the milk obtained from cows affected by clinical mastitis?*

*[**CMMILK**]_____*

32. Treatment protocol of clinical mastitis

32.1. *When do you treat clinical mastitis case? [**TCM**]* *1=At the first sign; 2=After a couple of days; 3=When it is severe; 4=What else_____?*

32.2. *What common treatment does the farmer follow*

*[**CTREAR**]?_____*

32.3. *What are the chosen antibiotics? [ANTIB]_____*

32.4*. What are the supportive drugs chosen?[**SDRUGS**]*

32.5*. Duration of treatment [**DURT**]_____*

32.6. *Any traditional treatment given*

*[**TTREAT**]_____*

32.7.*Who usually prescribes drugs*

*[**PRESCR**]_____*

33. What do you do with the milk that contains antibiotics?

[**CMMILK**]_____

B. Individual Cow Level Information (Animal Level Risk Factors)

1. Farm identity number [**FID**]_____/_____/_____/_____

2. Case/control animal identity number [**CASE/CONT**]___/___/___/____

3. Date of recording/interview [**DATE**]_____/_____/_____

4. Types of breed [**BREED**]_____

4.1. Breed blood percentage [**BBPERT**]_____

5. Age [**AGE**]_____

6. Parity [**PARITY**]_____

7. Body condition score [**BCS**]_____

8. Pregnant [**PREG**] 1=Yes; 2=No

9. Lactation stage [**LSTAGE**]_____

10. Daily milk yield before mastitis [**MYIELD**]_____

11. Suckled by calf **[SUCK]** 1=Yes; 2=No

12. Previous history of clinical masttis [**PCMH**] 1=Yes; 2=No

13. Was the cow treated with antibiotics before the last dry period? [**DRYT**] 1=Yes; 2=No

13.1. *If yes, how dry cow therapy is applied?*

*[**DTAPPL**]*_____

Observable characteristics of clinical mastitis:

No of quarters affected:_____

Mild (Grade-1): Abnormal milk: 1=Yes; 2=No

Moderate (Grade-2):Abnormal milk accompanied by swelling or redness of mammary gland:

1=Yes; 2=No

Severe (Grade-3): Exhibited systemic signs of illness such as depression, anorexia, dehydration, or fever:

1=Yes; 2=No

C. Quarter Level Information (Quarter Level Risk Factors)

1. Farm identity number [**FID**]_____/_____/_____/_____

2. Cow identity number [**CID**]_____/_____/_____/_____

3. Case/control quarter identity number [**CSQUART/COQUART**]___/___/___/____

4. Date of recording/interview [**DATE**]_____/_____/_____

5.Udder characteristics

5.1. *Udder symmetry [**USYM**]* *1=Yes;* *2=No*

5.2.*Teat shape [**TSHAP**] Conical/*_____

5.3. *Distance from teat to floor [**DTF**]*_____

5.4. *Distance between teats [**DTEATS**]*_____

5.5.*Visible lesions in teat (teat callosity)/udder[**ULESIONS**]* *1=Yes;* *2=No*

5.5.1. *If yes, what are those? [**TLESIONS**]*_____

6.Teat dipping

6.1. *Teat dipping is done [**TDIP**]* *1=Yes;* *2=No*

6.1.1*. If yes, pre and post dipping protocol is followed (before and after milking) [**DIPP**]*

1=Yes; *2=No*

6.1.2. *If yes, frequency of teat dipping [**FTDIP**]*_____

6.1.3. *Drying teat after washing [**DRYTW**]* *1=Yes;* *2=No*

Observable characteristics of clinical mastitis:

No of quarters affected:_____

Mild (Grade-1): Abnormal milk: 1=Yes; 2=No

Moderate (Grade-2):Abnormal milk accompanied by swelling or redness of mammary gland:

1=Yes; 2=No

Severe (Grade-3): Exhibited systemic signs of illness such as depression, anorexia, dehydration, or fever:

1=Yes; 2=No

D. Dynamic information

1. Farm identity number [**FID**]_____/_____/_____/_____

2. Date of visit and interview [**DATE**]_____/_____/_____ (**Vist**: 1/ 2/ 3/ 4/ 5/ 6)

3.Dynamic information about milking cows in the farm

[*event: Clinical mastitis/dried/culled/sold/slaughtered/died/stollen/any other loss]

No	Tag No	BCS	Age	Parity	Stage of lactation	Milk yield	Date of parturition/Date of calving	Date of entry into the study	Days of milking at entry into the study	Date of exit/*event into the study	Days of milking at exit/*event into the study

4. Dynamic information about pregnant heifer turned into milking cows in the farm

No	Tag No	BCS	Age	Parity	Stage of lactation	Date of parturition	Date of entry into the study	Days of milking at entry into the study	Date of exit/*event into the study	Days of milking at exit/*event into the study

5. Dynamic information about any new introducion of lactating cows (Dried to lactating or new purchased) to the farm

No	Tag No	BCS	Age	Parity	Stage of lactation	Milk yield	Date of parturition	Date of entry into the study	Days of milking at entry into the study	Date of exit/*event into the study	Days of milking at exit/*event into the study

6. Any changes in management (such as new milker, milking machine introduced ect.)

7. Climatic parametrs

7.1. *Ambient temperature [**ATEM**]_____ ℉*

7.2. *Humidity [**HUM**]_____*

7.3. *Rainfall [**RAIN**]_____*

7.4. *Radiation [**RAD**]_____*

8. If clinical mastitis

8.1. *Types of therapeutic treatments*

8.2. *Any traditional treatment given*

8.3. *Recovery rate*

8.4. *Who prescribes drugs?*

8.5. *Do you follow drug withdrawl period before milking for human/calf consumption?*

8.6. *Who prescribe drugs?*

Farm visiting schedule by Udder Health Team								
Sl No.	Name of the Farm	May	June	July	Aug	Sep	Oct	Day
1	Nahar Agro	10	14	12	9	13	11	Thursday
2	Desh Dairy	11	8	13	10	14	12	Friday
3	Amena Dairy Farm	1	1	3	7	4	2	Tuesday
4	Jaytun Dairy Farm							
5	Shorna Dairy Farm							
6	Dubash Agro	4	5	6	3	7	5	Friday
7	Bondhon Dairy Farm							
8	Shahnaz Dairy Farm							
9	Azizia Dairy Farm							
10	Star Dairy Farm							
11	Friendz Dairy Farm							
12	Bonoful Dairy Farm							
13	Ibrahim Dairy Farm	9	13	11	8	12	10	Wednesday
14	Mollah Dairy Farm	5	2	7	4	1	6	Saturday
15	Khamar Dairy Farm							
16	Moriyum Dairy Farm							
17	M M Agro Dairy Farm	12	9	14	11	8	13	Saturday
18	Jahan Dairy Farm							
19	Chowdhury Agro							
20	Dairy Fair	8	12	10	14	11	9	Tuesday
21	Homeland Dairy							
22	Bhuiya Dairy Farm							
23	Super Dairy Farm	7	4	2	6	3	1	Monday
24	Sabila Krisij Protisthan							

Appendix-II

Clinical Mastitis recording form

1.General Information

- **Farm name:**
- **Date of recording: _____/_____/_____**
- **Cow identification number:**
- Pregnant 1=Yes; 2=No
- Suckled by calf 1=Yes; 2=No
- Was the cow treated with antibiotics before the last dry period? 1=Yes; 2=No
- If yes, how dry cow therapy is applied? _____

2. Observable characteristics of clinical mastitis:

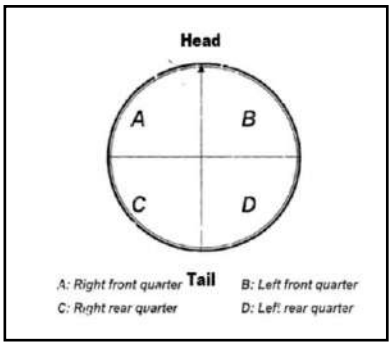
- **No of quarters affected: _____**
- **Mild (Grade-1)**: Abnormal milk: 1=Yes; 2=No
- **Moderate (Grade-2)**: Abnormal milk accompanied by swelling or redness of mammary gland:
  1=Yes; 2=No
- **Severe (Grade-3)**: Exhibited systemic signs of illness such as depression, anorexia, dehydration, or fever:
  1=Yes; 2=No

3. Udder characteristics

If any lesions, what are those? _____

4. Observable characteristics of clinical mastitis:

- **No of quarters affected: _____**
- Which quarter has been affected: **Left front/ Right Front/ Left hind/ Right Hind**
  :



- Name of drugs used: _____

Clinical Mastitis recording form (Bangla)

ওলান প্রদাহ নিবন্ধন ফর্ম

- ফার্মের নাম : _____
- তারিখ : ___/___/_____
- গরু চিহ্নিতকরণ নাম্বার : _____
- গর্ভবতী ? ☐ হ্যাঁ ☐ না
- বাছুর দিয়ে দুধ দোহানো হয় কিনা? ☐ হ্যাঁ ☐ না
- দুধ বন্ধ অবস্থায় এন্টিবায়োটিক্স দেওয়া হয়েছিল? ☐ হ্যাঁ ☐ না
- যদি দেওয়া হয় তাহলে এন্টিবায়োটিক্সের নাম ও ডোজ ? _____

ওলান প্রদাহের বিবরণ:

- মৃদু প্রদাহ (Grade 1): দুধে সমস্যা কিনা? ☐ হ্যাঁ ☐ না
- মাঝারি প্রদাহ (Grade 2): দুধ ও ওলানে সমস্যা: ☐ হ্যাঁ ☐ না
- তীব্র প্রদাহ (Grade 3): গরুতে দুধ, ওলান ও সামগ্রিক শারীরিক লক্ষণ (দূর্বলতা, খাওয়ার চাহিদা কম, জ্বর ইত্যাদি) ☐ হ্যাঁ ☐ না

আক্রান্ত বাটের বিবরণ:

- আক্রান্ত বাঁটের লক্ষণ? _____
- আক্রান্ত বাটের সংখ্যা : ☐ ১ ☐ ২ ☐ ৩ ☐ ৪
- কোন বাঁট আক্রান্ত চিহ্নিত করুণ :
- ব্যবহৃত এন্টিবায়োটিক্সের নাম: _____
- রোগের অবস্থা? (উন্নতি / কোন পরিবর্তন নেই/ অবনতি)

Grading of clinical mastitis

Grade of mastitis	Abnormal milk	Abnormal quarter	Sick cow
1 Mild	✓		
	Action: 1. Take milk sample. 2. Record information		
2 Moderate	✓	✓	
	Action: 1. Take milk sample. 2. Record information		
3 Severe	✓	✓	✓
	Action: 1. Treat her immediately.		

Appendix-III

Sample collection procedure

Required materials for milk sample collection

- Sterile vials/tubes
- 70% Alcohol
- Cotton for alcohol soak
- Cooler ice box
- Tube/vial holding rack
- Disinfectant for cleaning teats (70% Alcohol, 0.5% w/v Chlorhexidine)
- Tissue papers for individual teat preparation
- Permanent ink pen, gloves and mask.

Procedure

1. Labeling tubes using permanent ink pen with Date, Farm, Cow, Quarter
2. Removal of loose dirt, bedding and hair from the teats and wash the teats and udder
3. Discard 1-2 streams of fore milk from affected quarter
4. Dipping/wiping all quarters with disinfectant solution keeping at least 30 seconds in cotact with teats
5. Dry the individual affected teat
6. A single alcohol soaked cotton is used for each individual teat prior to sample collection and not to touch the clean teat ends
7. The test tube is held at 45° angle in left hand
8. Sample should be collected closest teat to far teat
9. Collection of 5-10mL of milk
10. Disinfect the affected quarter
11. Store the milk sample vial with rack in ice box with ice pack.

Milk Sample Collection Procedure in Picture

Marking the test tube

Clean and dry udder

Fore-strip to discard few drops

Wipe with antiseptics

Take two squirts of milk

Transport of ice box

Collection of Bulk Milk for Bulk milk somatic cell count (BMSCC)

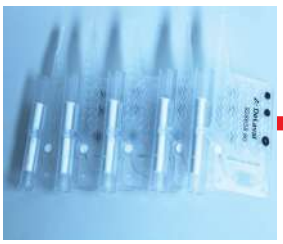
Composite milk in Tank

Thoroughly mixing of the milk

Bulk milk sample collection

Milk sample in collection tube

DeLaval somatic cell counter



Cassette

Somatic cell counting

Recording the results

Ref: National mastitis council (http://www.nmconline.org/wp-content/uploads/2016/09/Procedures-for-Collecting-Milk-Samples.pdf)

Appendix-IV

3.11.1.1. Blood agar inoculation

The milk sample stored at -20°C was thawed at room temperature and inoculated into blood agar (Oxoid®, Oxoid Ltd, Basingstoke, United Kingdom). No growth in first inoculation was rechecked using the second inoculation in blood agar. No growth indicates absence of bacteria in the collected milk sample. If growth of colonies were observed, color, size and appearance of colonies were recorded for each sample. Growth of three or more types of colonies in a blood agar was suspected as contamination (Pinho et al., 2008; Persson et al., 2011). If two different colonies were found, they were then separately inoculated in Mannitol Salt Agar (MSA) and MacConkey Agar (MAC) for differentiating the organisms.

3.11.1.2. Mannitol Salt Agar inoculation

Samples properly grown in blood agar were further inoculated into MSA (Oxoid®, Oxoid Ltd, Basingstoke, United Kingdom) and the positive growth in MSA indicates the presence of *Staphylococcus* spp. (De Visscher et al., 2013). MSA Positive samples were then undergone for coagulase test and catalase test. No growth was revealed in MSA or MAC but growth was observed in blood agar (write growth characteristics here) were suspected as *Streptococcus* spp. (Persson et al., 2011).

3.11.1.3. Catalase test

Catalase test detects the presence of catalase enzyme in the bacteria. Some bacteria has catalase enzyme which help them escaping from oxidative damage by Hydrogen peroxide (H_2O_2). Catalase enzyme breaks down the H_2O_2 into water and bubble which is visible by rapid formation of bubbles (Cowan and Steel, 1965). This is useful for differentiating catalase negative bacteria like *Streptococcus* spp. from catalase positive bacteria such as *Staphyococcus* spp., *Micrococcus* spp. as well as *Bacillus* spp.

3.11.1.4. Coagulase test

Coagulase positive *Staphylococcus* spp. can stimulate clotting of plasma using coagulase enzyme which can be detected using coagulase test (Foster and Geoghegan, 2015; Bonar et al., 2018). Negative results in coagulase test but positive in MSA were suspected as coagulase negative *Staphylococcus* spp. (Persson et al., 2011).

3.11.1.5. Gram staining

Unknown/unidentified colonies in MSA were subjected to catalase and coagulase test as well as Gram's staining to identify gram positive and gram negative bacteria (Panawala, 2017).

3.11.1.6. Gram-negative bacteria isolation

Gram negative bacteria was isolated by the following methods:

3.11.1.6.1. MacConkey agar inoculation

All samples grown in blood agar were inoculated in MAC agar (Oxoid®, Oxoid Ltd, Basingstoke, United Kingdom) for isolation of all gram negative bacteria. In MAC agar, two types of colonies were observed; pink color colonies for lactose fermentative bacteria and colorless colonies for non-lactose fermentative bacteria (Persson et al., 2011).

3.11.1.6.2. Eosine Methyline blue agar inoculation

Positive growth with pink colored colony in MAC agar was inoculated in EMB agar (Oxoid®, Oxoid Ltd, Basingstoke, United Kingdom) for confirmation of *E. Coli*. Greenish metallic sheen was characteristic to *E. Coli (McVey and Czuprynski, 2013).*

3.11.1.6.3. Biochemical tests

Further confirmation of gram negative species is supported by some specific biochemical tests. (Garcia, 2010; Murray et al., 2015; Wanger et al., 2017)

Indole test

Pink colored ring formation in the test tube within 30 seconds indicated positive test. This was characteristic for Enterobacteriaceae. Kovac's reagent gives more sensitive results for *E. coli*. that gives characteristic colony in EMB agar (Murray et al., 2015)**.**

Citrate utilization test

Fermentation of citrate and change of color from greenish to prussian blue to yellow indicated positive and changing of no color indicated as negative. This test gives useful evidence for differentiation between *Citrobacter freundii* and *Escherichia coli (Garcia, 2010)*.

Motility test

Diffuse or hazy growth in SIM (Sulfide, Indole, Motility) media indicated motile bacteria. Enterobacteriaceae can be differentiated using motility test (Shields and Cathcart, 2011).

Oxidase test

Oxidase reagent in presence of cytochrome oxidase containing bacteria produced blue color in filter paper. Enterobacteriaceae give negative results but Pseudomonadaceae give positive result for oxidase test (Shields and Cathcart, 2010).

Methyl Red test

Formation of red color within few minutes indicates positive test results. *E. coli* is positive but *Enterobacter* gives negative results (McDevitt, 2009).

Voges-Proskauer (VP) test

Development of red color after 15 minutes or more indicated positive results. *Klebsiella* spp. *and Enterobackter gives positive* results while *E. coli, Citrobacter* spp. and *Pseudomonas* spp. gives negative results for VP test (McDevitt, 2009).

3.8.1.3 Maintenance of pure culture and stock

For isolation of pure culture the bacteria are grown in selective media: MSA or in MAC agar was again sub-cultured in blood agar. Those bacteria had not grown in MacConkey agar or Mannitol salt agar but previously had grown in blood agar inoculation, from sample they were again sub-cultured in blood agar. After confirmation of pure culture by observation of similar type colonies in sub-cultures in blood agar, colonies were reinoculated into brain heart infusion broth and incubated at -37°C for 24 hours for bacterial multiplication as per manufacturer instruction (Oxoid®, Oxoid Ltd, Basingstoke, United Kingdom). 50% buffered glycerol was prepared through mixing of 50 parts of buffered glycerol with 50 parts of normal saline. Then 700µl overnight cultures were transferred in sterilized cryovial with 300µl of 50% buffered glycerol at the temperature of -80°C as stock for longer time preservation (Petti and Carroll, 2011). Entire procedure for BC has been attach as sketch below (Figure 1).

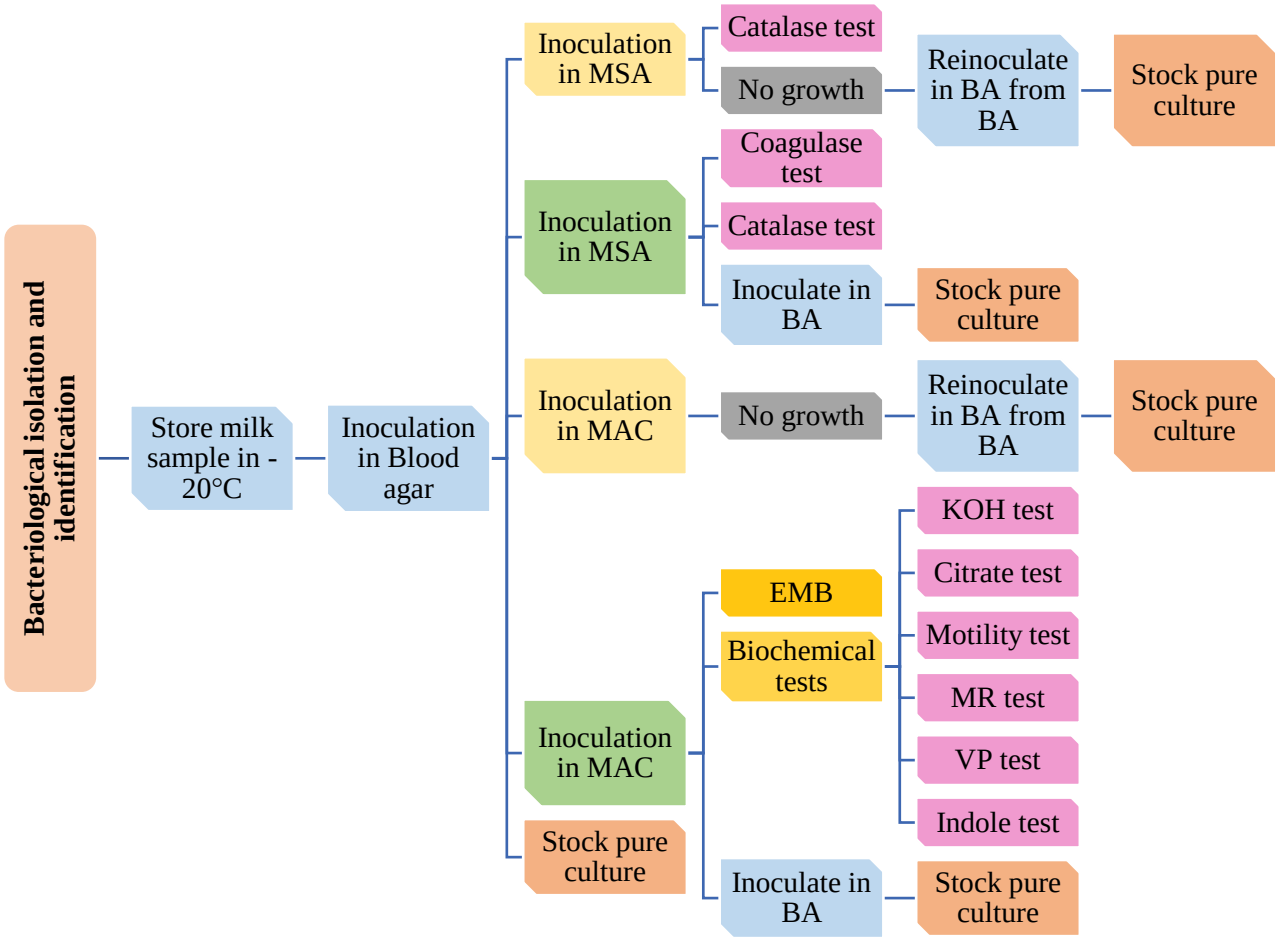
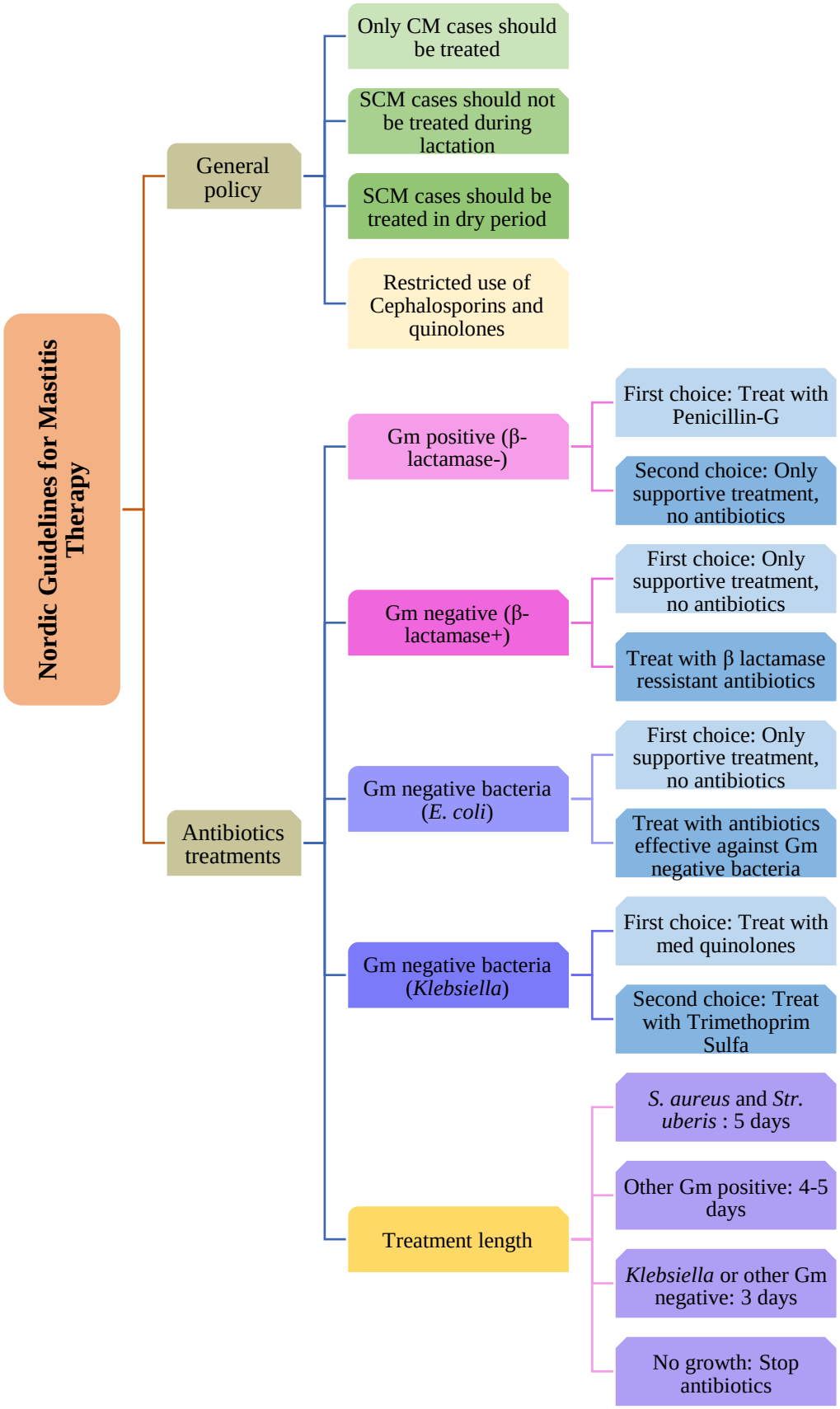


Figure 1. Flow chart for bacteriological isolation and identification

Appendix-V

Nordic Guidelines for Mastitis Therapy